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141st Master's Thesis

Thesis Advisor: Prof. Chansik Park

**Personalized Construction Safety
Training System using Conversational
AI-based Virtual Reality**

August 2024

**The Graduate School
Chung-Ang University
Department of Computer Science and Engineering
Major in Application Software
Aqsa Sabir**

Personalized Construction Safety Training System using Conversational AI-based Virtual Reality

**Presented to the Faculties of the Chung-Ang University in
Partial Fulfillment of the Requirement of the Master's Degree**

August 2024

**The Graduate School
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**Personalized Construction Safety Training System using Conversational
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ABSTRACT

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Construction safety training is crucial for equipping workers with the necessary knowledge and skills to mitigate potential hazards on job sites. Traditional one-size-fits-for-all training methods have often proven inadequate in effectively preparing the workforce for the dynamic challenges encountered in the field. This research explores the integration of virtual reality (VR) technology and artificial intelligence (AI), specifically large language models (LLMs), to enhance personalized safety training for the construction industry. The proposed system combines immersive VR environments with an LLM-based intelligent safety trainer (iSafeTrainer), enabling trainees to engage in conversational interactions tailored to relevant trade such as electrical, scaffolding, or welding with novice, beginner, or expert knowledge level. Through this personalized approach, the system aims to improve safety knowledge among construction workers.

To evaluate the effectiveness of the iSafeTrainer-based VR system, an experimental study was conducted involving questionnaire-based surveys.

Participants interacted with the VR system and engaged with the LLM-based iSafeTrainer, providing feedback through a combination of structured and unstructured questions. The evaluation of the proposed system has been done using 5 key factors 1) user satisfaction and overall usability 2) user experience and interaction 3) training effectiveness and engagement 4) system guidance and support 5) impact on confidence and preparedness. The evaluation results indicate effective results and participants highlighted the advantages of the iSafeTrainer over traditional methods, particularly in terms of tailored content delivery, immersive virtual environments, and real-time guidance and feedback from the AI-powered iSafeTrainer.

This research contributes to the advancement of construction safety training by demonstrating the potential of integrating VR and AI technologies to create personalized, interactive, and engaging learning experiences. The iSafeTrainer system showcases the feasibility of adapting training content dynamically based on individual needs, addressing a crucial gap in existing safety training approaches.

Keywords: Construction Safety, Personalized Safety Training, Conversational-AI, Virtual Reality

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1. Introduction

The construction industry is known for its complexities and dynamic nature, which presents significant safety challenges related to dynamic work environment, low budgets, technology, lack of safety knowledge, and physical demands [1]. These challenges are particularly prominent for construction students and workers, who may lack the necessary safety knowledge and experience due to their newness to the field. Therefore, efficient, and effective safety training methods are imperative for these individuals [2]. Labor costs, which constitute a significant portion of total project expenses, are directly impacted by the skill level and safety knowledge of the workforce [3]. Consequently, unskilled labor, influenced by low profit margins, can significantly contribute to safety incidents, emphasizing the imperative need for safety training [3], [4]. But traditional lecture-based classroom safety training has frequently proven insufficient in adequately equipping workers to address the distinct challenges encountered on-site [5]–[7]. These traditional training approach typically involves conveying information through text, images, and videos with the aim to impart essential safety knowledge and practices to the workforce [8]. As the construction industry evolves, new training methods are rising, emphasizing visualization technology. This innovative visualization technology approach utilizes Virtual Reality (VR)

environments, simulated videos, and text-based instructions within VR for a more immersive learning experience [9]–[11].

Both traditional lecture-based training and visualization technology training prioritize customized content, to address learning needs of students and workers [12]. Customized content in both traditional lecture-based and visualization technology training typically takes the form of pre-defined courses and pre-plotted VR tasks, reflecting the industry's acknowledgment of the necessity to adapt training materials to workforce requirements [13], [14]. Literature highlights that predefined simulated tasks in visualization technology are established because of field-specific training, often referred to as personalized safety training. That's why, the importance of personalized content in safety training, particularly for students and workers, has been extensively explored in literature [15]–[17]. Furthermore, existing studies have highlighted the benefits of tailored safety training methods, such as creating modified VR tasks and utilizing Non-Playing Characters (NPCs) to deliver instructions in VR environments. These approaches emphasize interactive and customized individual directives training, showcasing the potential to enhance safety knowledge acquisition and retention among learners [17], [18].

Further, integrating visualization technology such as Virtual Reality (VR) with Artificial Intelligence (AI) presents a cutting-edge method for efficiently delivering personalized safety training, showing promise in effectiveness [19]. VR has demonstrated potential as a substitute for real-world task training,

particularly in hazardous scenarios, due to its immersive nature and ability to provide real-time feedback. However, traditional computer inputs may hinder the effectiveness of VR training, especially for workers with limited skills and literacy [20]. AI, particularly Large Language Models (LLMs) like GPT-4, can address this limitation [19]. These models excel in understanding and generating human-like text, making them ideal for creating conversational agents or chatbots. By integrating AI-based conversational agents into VR training systems, workers can ask questions and receive instant feedback in a conversational setting, enhancing engagement and understanding while reducing cognitive burden [21], [22]. However, there remained a gap between the capabilities of an individual and specific requirements-based training in a conversational setup, this gap needs to be addressed of continually modifying VR environments and training AI NPCs repeatedly for personalized training.

1.1. iSafeTrainer based VR System

This study develops a system by integrating Large Language Models (LLMs) into VR environments to enhance personalized safety training for construction students and workers. By combining LLM technology with VR, tailored guidance within training scenarios is offered, aiming to improve personalized and user specific training. Unlike prior research, which relied on manually designed scenarios, this study utilizes LLMs to deliver personalized safety training based on individual trade and knowledge levels in a conversational setup. The proposed study involves gathering user personal

information and transmitting it to the AI system. This process aims to create a personalized LLM-based NPC for the needs of a user within a relevant virtual reality environment. The system covers electrical, scaffolding, and welding safety training across novice, beginner, and expert levels of knowledge. The focus on electrical, scaffolding, and welding safety training is well-justified due to their high risk in the construction industry. A study by the Occupational Safety and Health Administration (OSHA) highlights these trades as major contributors to workplace fatalities and serious injuries. By targeting these high-risk areas, the author addresses critical safety concerns in construction [23]. The decision to cover safety training across novice, beginner, and expert levels is supported by research that emphasizes the need for tailored training to address varying levels of experience and expertise among construction workers [24]. Novice workers often lack foundational safety knowledge, while beginner workers need additional guidance. Expert workers benefit from advanced training to stay current with safety practices and regulations.

By catering to these distinct knowledge levels, the proposed VR system ensures comprehensive safety training that meets the diverse needs of the construction workforce. LLM is integrated into VR simulations, allowing trainees to interact with an intelligent safety trainer (iSafeTrainer) in a conversational setting based on their personalized directions. The effectiveness and evaluation of the system are conducted through questionnaire-based surveys, which involves providing paper-based questions to students or

workers who communicate and train with the iSafeTrainer from the proposed system for personalized communication and training. Questionnaire-based surveys are an important source of information for evidence-based research. They embody self-reports and therefore gather the participants' subjective experiences, question types in surveys divided into structured and unstructured questions, structured questions allow for a clear classification of the responses, while unstructured or open-ended questions allow the subjects to respond freely [25]. This assessment is rooted in effectiveness of the proposed system for individuals, focusing on their ability to acquire and retain new information, skills, or competencies through participation in the surveys.

1.2. Contributions

The major contributions of this research include:

- Development of virtual construction sites with interactive VR interfaces for safety training, focusing on common hazards.
- Introduction of an LLM-based iSafeTrainer tailored to individual directives to meet specific training goals.
- Integration of content delivery via communication with the iSafeTrainer to achieve targeted objectives and training.
- Validation of the efficacy of LLM-based iSafeTrainer VR training scenarios through questionnaire-based surveys.

The remainder of this paper follows this structure: Firstly, it discusses the challenges of traditional construction safety training methods and proposes solutions suggested in existing literature. Next, it details the research methodology, including system architecture and development, with subsections on gaming platform development and practical implementation. Following this, an experimental case study on personalized safety training is presented, along with an examination of the evaluation results are then discussed and interpreted. Furthermore, a thorough discussion on the system's effectiveness, results and improvements has been presented. Finally, the paper concludes by summarizing its contributions, limitations, and future work.

2. Literature Review

2.1. Safety Training in the Construction Industry: Traditional Methods vs. Visualization Technologies

Traditional Methods: Safety training in the construction industry is essential due to inherent risks [1]. Mohammed et al. and Boateng et al. observed that traditional safety training for construction students mainly involves conventional methods, such as lectures with texts and images. [5], [8]. This method is further supported by studies [26]–[28], which describe additional techniques like toolbox talks, job demonstrations, and the use of printed materials, including manuals, posters, and videos. While Beş et al. acknowledge that these methods are familiar and easy to implement for both trainers and trainees, they also note that such approaches can be time-consuming and disrupt workflow [29]. Additionally, these materials may not address specific job site hazards or individual learning preferences, potentially limiting their effectiveness.

Furthermore, the critique by Rafindadi et al. highlights a significant drawback of traditional methods, their limited ability to actively engage learners and accurately reflect the unpredictable nature of construction sites [30]. This gap between theoretical knowledge and on-site practical application is problematic, potentially leaving workers unprepared for real-world scenarios. This issue is particularly acute for new student workers, as noted by Åsgård et

al. [31]. Liu et al. further emphasize that construction workers struggle with comprehending standardized safety concepts in a classroom setting, often due to limited experience [32]. This struggle is compounded when applying theoretical knowledge in practical situations, where quick decision-making and situational awareness are crucial, as observed by Kaplan et al. and Hussain et. al. [33], [34].

In response to these challenges, recent literature has explored the incorporation of visualization technologies, such as virtual reality (VR), augmented reality (AR), and Building Information Modeling (BIM), as innovative means to enhance safety training. Kaplan et al. highlighted the potential of these technologies to create immersive and interactive learning environments, simulating real-world construction scenarios [33]. This approach, further explored by Akinlolu and Haupt, and Afzal et al., suggests a promising avenue to bridge the gap between theoretical knowledge and practical application, potentially leading to improved safety outcomes on construction sites [35], [36].

Visualization Technologies Approach: These advanced technologies were further explained by Zhu et al. that how virtual reality (VR) and augmented reality (AR) technologies create immersive environments where workers can simulate hazardous scenarios without real-world risks [37]. Additionally, alone VR gained much attention and many researchers explored the use of VR and the limitation associated with these studies [14], [38]. Akinlolu and Haupt,

along with Jelonek et al., have further expanded on this by highlighting the effectiveness of BIM-based visualization and simulation software in creating realistic training scenarios, enhancing the understanding and management of construction site health and safety [35], [39]. However, these advanced methods are not without limitations. Li et al. noted a key challenge which is the use of pre-plotted or pre-set VR tasks and one-size-fits-for-all training content [14]. This approach may not adequately cater to the diverse learning needs and preferences of students and workers. Despite this, advanced technologies hold a core advantage in their ability to offer immersive, personalized learning experiences in the form of pre-plotted and customized VR tasks. Xu et al. and Lee et al. underscored the potential of these technologies in providing tailored training experiences that can dynamically adjust to the progress of individual learners [40], [41]. Therefore, it is crucial to develop personalized and adaptive training programs with real-time feedback and guidance, that should be capable of dynamically adjusting to the progress and requirements of individual learners, thereby enhancing the effectiveness of safety training.

In response, recent researchers have investigated integrating AI, especially LLMs, into VR environments [42]–[45]. Several survey studies explore the use of LLM-based agents in training environments and highlight the associated limitations [46]–[49]. Moreover, The application of LLM-based NPCs in VR, as discussed by Auxtero, Anand, Bozkir, and Sweetser, shows potential for

improving interactivity and personalization in learning [42], [50]–[52]. Such advancements indicate a growing trend towards more tailored, communication-focused training methods, aiming to provide more effective and relevant training experiences that align closely with the real-world demands of the construction industry.

2.2. Personalized Safety Training with VR and AI

Integrating virtual reality (VR) with artificial intelligence (AI), particularly leveraging large language models (LLMs) and non-player characters (NPCs), offers a promising avenue for safety training. Studies by Jeelani et al. and Xu & Zheng focus on utilizing VR for creating immersive and interactive training environments. Jeelani et al. emphasize personalized safety protocols using VR and stereo-panoramic environments, highlighting the need for further research on user engagement in diverse real-world scenarios [53]. Xu & Zheng present a multi-level VR-based training system for urban construction sites, but they note challenges in hardware integration and simulating dynamic site conditions [54]. Cruz et. al. provides a comprehensive review of VR in construction safety, identifying the potential of VR in training and its limitations, particularly the gap between VR experiences and real-world application [55]. Eiris et al. compare 360-degree panorama and VR techniques, discussing their respective advantages and limitations in user experience and technology accessibility [56].

Further studies such as Joshi et al. explore VR for safety training in the concrete industry, acknowledging issues in technological adoption and simulating complex industry scenarios [57]. Kaarlela et al. investigate the use of digital twins and VR for safety training, facing limitations regarding VR technology's maturity level and integration with digital twin models [58]. A research study by Li et. al. focuses on synthesizing personalized VR training scenarios, confronting challenges such as the high dependency on advanced VR technology and the creation of accurately reflective real-life scenarios [14]. Wen et. al. described virtual humans in VR for communication training in AEC education (2023) show promise in enhancing interaction skills but struggle with replicating complex human interactions [38]. Furthermore, a research study by Hussain et. al. on an AI-based VR system for migrant worker training highlights the effectiveness of AI in overcoming language barriers, yet it points out the complexities of programming AI for diverse linguistic and cultural backgrounds [19]. Additionally, Wen and Gheisari explore the use of virtual humans for communication skills training in 360-degree virtual field trips, it faces challenges in accurately simulating the nuances and complexities of real-world human interactions within the virtual environment [38]. Below is Table 2-1 summarizing key research studies integrating VR, AI, and LLM-based NPCs and for safety training.

Table 2-1: Summary of Key Research Studies Integrating VR, AI, and LLM-based NPs.

Year	Author	Objective	Technology	Limitation	Reference
2022	Li et al.	Synthesizing personalized VR training scenarios	VR	High dependency on advanced VR technology i.e., FE-VR and PG-VR	[14]
2023	A. Anand	Focus on ChatGPT in Virtual NPC Interactions	AI, NPCs and	No data related to real-world application	[59]
2023	Y. Ge, et al.	Envisioning AIOS, agents and the AIOS-agent ecosystem	LLM-based AI Agent Applications	Dependency on pre-trained systems	[43]
2023	D. Chen, et al.	Multi-agent collaborative framework for game development	AI NPCs and	The lack of support for multi-party, multi-modal interactions	[44]
2023	M. Klissarov, et al.	Studying AI feedback and intrinsic motivation	AI, LLM	Lack of context, common sense, and emotional intelligence, as well as ethical concerns	[45]

2023	Wen et. al.	Communication training in AEC education using virtual humans	VR	Struggle with replicating complex human interactions	[38]
2023	T. Händler	Taxonomy for autonomous LLM-powered multi-agent architectures	Autonomous LLM-powered agents	Systematic analysis for engineer and AI practitioner use should be clearly identified and discussed	[46]
2024	S. Hu, et al.	Survey on conceptual architecture of Large Language Model-Based Game Agents	LLM-based game agents	Lack of in-depth analysis, and dependence on the quality and availability of data	[47]
2024	Q. Ma, et al.	Survey on autonomous LLM-powered multi-agent architectures	Autonomous LLM-based agents	Balancing autonomy and alignment challenges	[48]
2024	F. Li, et al.	Developing LLMs for Behavior Tree Generation	LLM-based agents	Construction of effective fine-tuning	[49]
2024	Hussain et. al.	AI-based VR system for migrant worker training	AI and VR	Complexities of prompt engineering AI for diverse linguistic and cultural backgrounds	[19]

Reviewing the research reveals a significant limitation in current personalized safety training methods using advanced tools like VR and AI in the construction industry. The primary issue lies in the necessity for individualized directives-based training (i.e., trade, level of knowledge or experience level etc.) to meet the complexities and requirements of construction students and workers as Hussain et. al. addressed the need by focusing on the language barrier faced by migrant workers [19]. Further, the methods described in literature typically depend on pre-planned tasks, lacking the dynamic interaction and tailored feedback necessary to address the specific requirements of each trainee. This shortfall is crucial in construction, where the precision and relevance of safety training directly impact real-world situations and individual safety. To address this gap, the proposed approach aims to improve construction safety training by developing personalized, communication-focused, content delivery methods. Tailored to each trainee's specific needs, it promises more effective and relevant training, enhancing understanding and practical application of safety protocols in real-world construction scenarios. This approach signifies a notable advancement by offering an intelligent LLM-based iSafeTrainer for personalized and adaptive communication-based training. This is particularly valuable due to the varied requirements within the construction workforce.

2.3. Motivation

The motivation for this research originates from the promising potential of applying constructivist approaches to personalized safety training for construction students and workers. Constructivism, a well-established learning theory, highlights four key elements such as situation, collaboration, conversation, and meaning construction [16]. It underscores the active process of learners constructing meaningful knowledge and fostering critical thinking through engagement with their environment. This theory suggests that learners achieve a deeper understanding of environment when they participate in the learning process, building knowledge through direct experiences and interactions, while the instructor serves as a mentor [60], [61]. Wenjuan et. al. leverages VR technology, constructivism, and situated cognition to develop a personalized and effective learning framework for safety-relevant applications. Bęś et. al. argued that constructivism fosters critical thinking, cultural awareness, and engagement, which are essential for effective hazard identification and risk perception among construction student workers [62].

The proposed model and methods seek to enhance and improve the transfer of learned skills to real-world safety situations [63]. It seeks to apply the above-mentioned principles to safety training by integrating LLMs into immersive VR environments, thereby offering tailored, interactive training scenarios. This modern approach aims to enhance knowledge gain among construction workers by providing training that aligns with their specific trades and

individual knowledge levels. By enabling trainees to engage with an intelligent safety trainer in a conversational setting, the study promotes an interactive and adaptive learning experience, mirroring constructivist ideals of learner-centered education. This personalized and engaging method of training not only has the potential to improve safety knowledge but also supports the development of critical thinking and problem-solving skills crucial in real-world construction settings. Figure 2-1 illustrates the Venn diagram that describes the educational process characterized by a dynamic interplay of several core elements. The learning environment sets the stage, providing a nurturing and responsive setting that encourages exploration and the construction of knowledge. Assessment within this framework is formative, continuous, and seamlessly integrated into the learning trajectory, serving as a reflective tool for both educator and learner. Engagement is paramount and is fostered through a multi-faceted approach, ensuring that learners are emotionally, cognitively, and behaviorally invested in the educational experience in the presence of iSafeTrainer. Content delivery becomes more experiential, in the proposed environment learners are not only receiving information but are also experiencing it. For instance, instead of reading about construction hazards, they might interact with them in an environment to identify them which often employs critical thinking exercises that compel active participation. Interactions enhance learning, offering diverse viewpoints and collaborative discourse, essential for a deeper understanding. The role of

the teacher (iSafeTrainer) is reimagined as a guiding presence, offering feedback that is less corrective and more reflective, facilitating the learner's personal journey of knowledge construction. At the convergence of these elements lies the individual's directive learning, representing the active role of player/learner in synthesizing new information with existing knowledge is at the heart of this approach, ensuring that learning is personalized and meaningful.

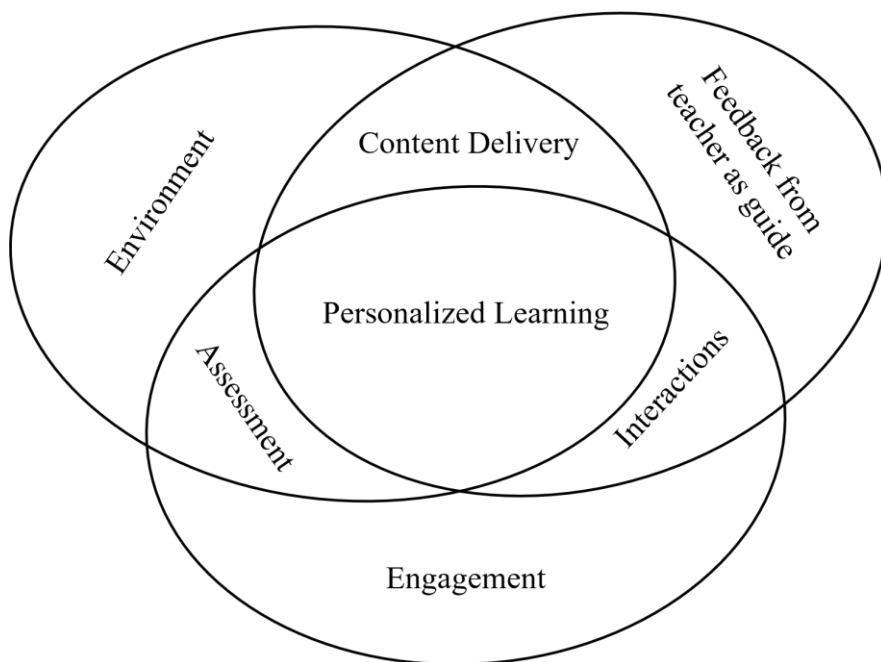


Figure 2-1. Constructivist Learning Principles with respect to Personalized Safety Training

3. Research Methodology

This section outlines the research methodology of the proposed VR system (Figure 3-1). Initially, an introduction reviews traditional safety training, discusses visualization technologies and associated limitations, and introduces the proposed VR system as an advanced solution. A comprehensive literature review details various safety training methods, their limitations, and a strong motivation for the proposed system which also explores the link between constructivism and personalized safety training. The methodology is divided into two parts: the system architecture and development, and the practical implementation of the proposed VR system. Special emphasis is given to the LLM-based iSafeTrainer component, which uses AI prompt engineering to customize interactions based on user selections from the main menu. This feature allows iSafeTrainer to recommend video-based content within the VR environment for more effective learning and hazard mitigation. A case study illustrates functionality of proposed VR system and the role of iSafeTrainer, with experiments to assess the effectiveness of the system in enhancing safety knowledge among students and workers. Evaluation results of the proposed system have been done after experiments. Next section concludes with a discussion of the contributions and potential limitations of the system, suggesting directions for future research.

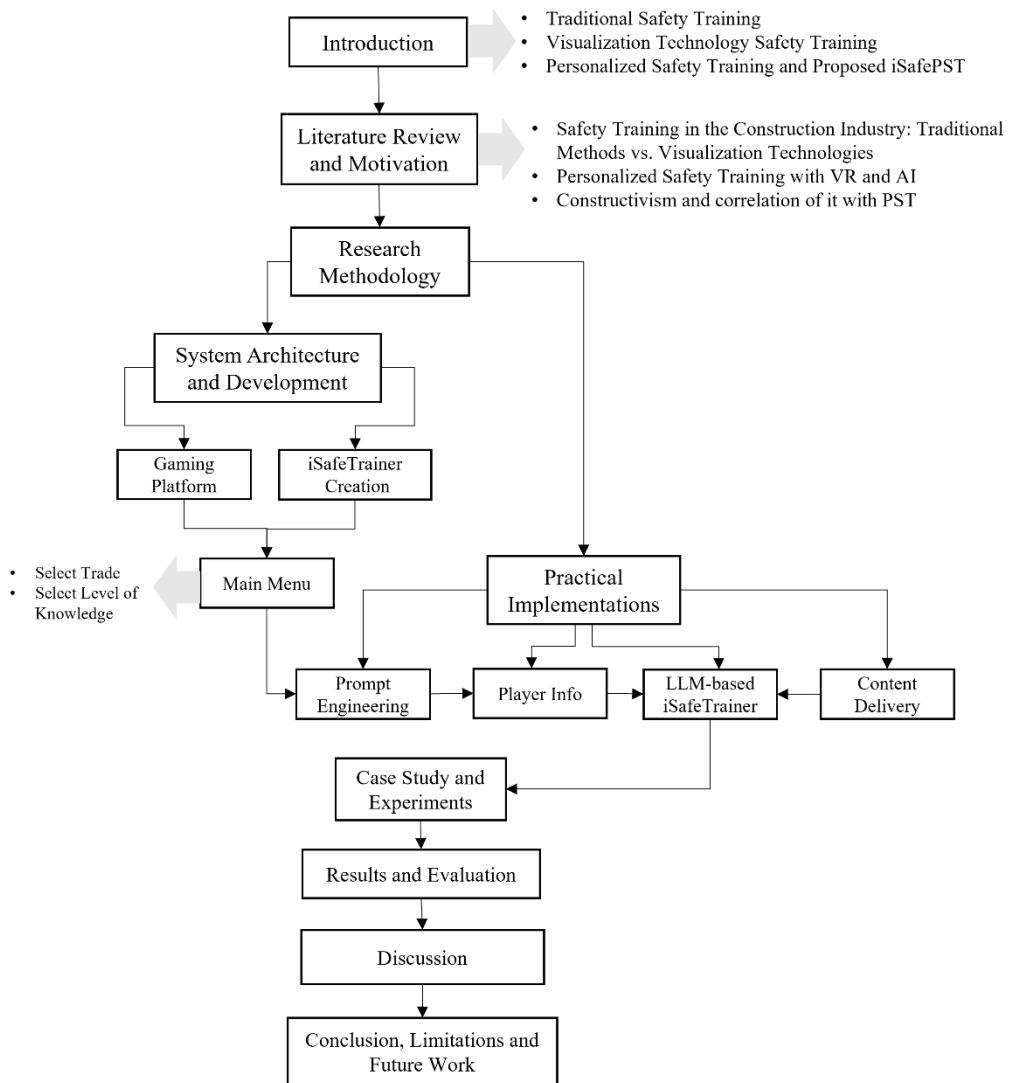


Figure 3-1. Research Methodology Framework for the iSafeTrainer based VR system.

3.1. System Architecture and Development

An experimental approach was used as the research methodology in the current study to meet the research objective. As discussed, the study leveraged constructivist learning theory, to satisfy the personalized safety training needs of students and workers by providing an authentic, relevant, and applicable

learning environment tailored to their distinct learning and work contexts as shown in Figure 3-2. This theory emphasizes active learning, where individuals construct their own understanding and knowledge through experiences, interactions, and communication with the instructor and environment [61]. In line with this theoretical framework, a contextualized version of an LLM-based virtual instructor as iSafeTrainer was integrated into the VR environment to enhance the learning experience. Figure 3-2 also emphasized that students and workers can read, visualize, perform in the VR environment, and can have communication with iSafeTrainer.

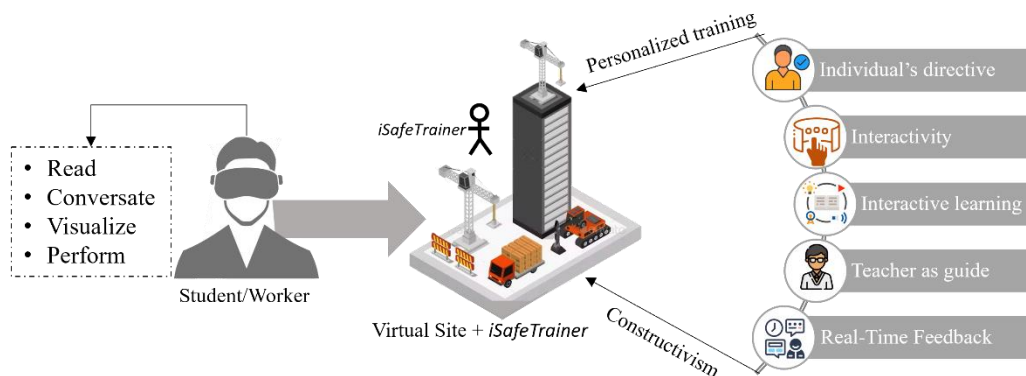


Figure 3-2. Correlation of Constructivism and Personalized Safety Training

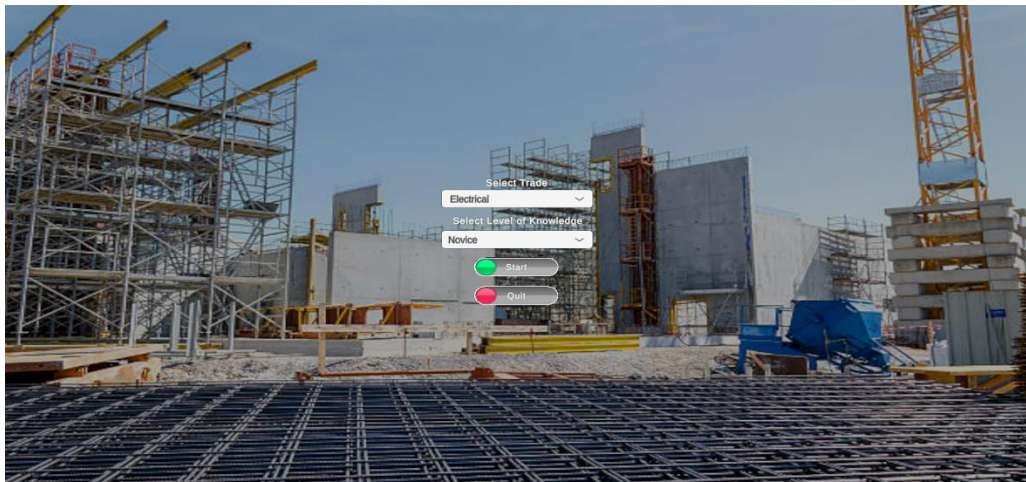
3.1.1. Gaming Platform Development

The gaming platform was developed using unity 3D game engine on a Windows 11 system equipped with an Intel Core i9 tenth- generation processor operating at 3.70 GHz, complemented by 16GB of RAM. The version 2021.3.16f1 of unity editor was used to develop proposed iSafeTrainer system.

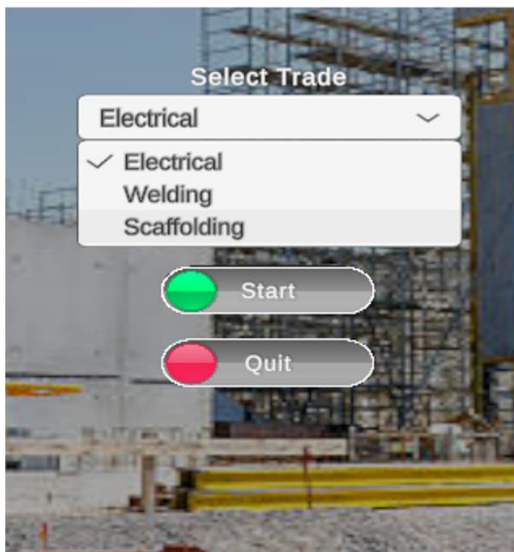
Unity-based Construction Site: The system architecture was developed using the Unity game engine to create an immersive virtual reality training experience. This was achieved by integrating two crucial components: a three-dimensional (3D) mesh obtained from an actual construction site and custom-built virtual 3D objects. The 3D mesh was generated using Gaussian Splatting to capture the authentic construction environment, augmented by a pre-existing 3D mesh sourced from the open-source Sketchfab web platform. This combination provided a solid foundation, delivering a lifelike representation of a construction site within the virtual environment [64], [65]. Furthermore, virtual 3D objects, including buildings, construction equipment related to electrical, scaffolding, and welding hazards, were either acquired from the Unity Assets Store or created using Blender 3.5 software. To enhance realism, realistic sounds of construction machinery and site noises were simulated, replicating the auditory atmosphere of a genuine construction site.

Menu UI and Training Platforms: The initial interface presents a main menu, allowing users to choose their trade and level of knowledge such as Novice, Beginner, or Expert from Electrical Safety, Scaffold Safety, or Welding Safety Training. Users can begin their training by selecting 'Start', or exit using the 'Quit' button as shown in Figure 3-3. Based on user selections, individuals are guided to specific environments where hazards are marked with yellow arrows. A proficient iSafeTrainer is available for interaction. Hazards are highlighted to ensure that students and workers can identify and discuss them with

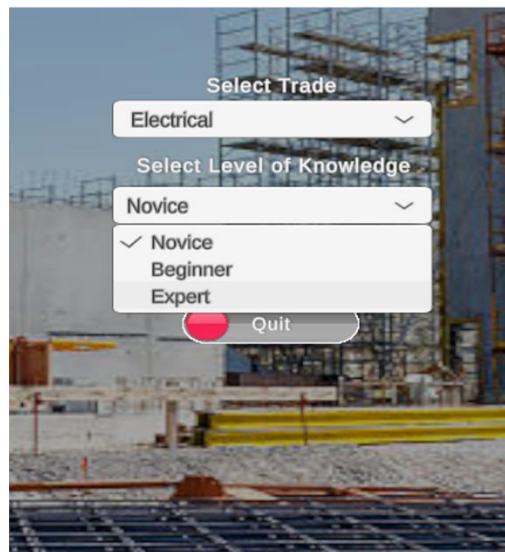
iSafeTrainer. If any hazards are overlooked, iSafeTrainer will ask the users to revisit and identify the remaining hazards, serving as an assessment of their knowledge gain and active learning.



(a)



(b)



(c)

Figure 3-3. (a) Main Menu Structure; (b) Sub-Menu for Trade Selection; (c) Sub-Menu for Level of Knowledge Selection

Electrical Safety Training Platform: This study develops a VR-based electrical construction site, providing a realistic environment for users to identify various hazards. The immersive VR setup invites participants to explore the site, engaging them in observational and critical thinking activities to understand the impact of hazards on worker safety through learning content and discussion with iSafeTrainer. Key hazards include wet floors, misplaced overhead power lines, and exposed wires, but safety cones, fire extinguishers, distribution boxes to confuse the players so that they can distinguish between hazards and hazard mitigation equipment's. The design of this VR environment focuses on enhancing user engagement and the development of hazard recognition skills, as shown in Figure 3-4.



Figure 3-4. Hazardous Area of Electrical Safety Training VR Environment

Welding Safety Training Platform: Focused on welding safety training, this platform immerses users in realistic scenarios to understand and identify hazards. Grounded in constructive learning theory, it is particularly beneficial for construction students and workers. Users navigate a virtual jobsite, with highlighted hazards like fire blast because of sparks on oil cane, electrical shock, physical hazards, and the dangers of not wearing proper protective equipment (PPEs) as illustrated in Figure 3-5.

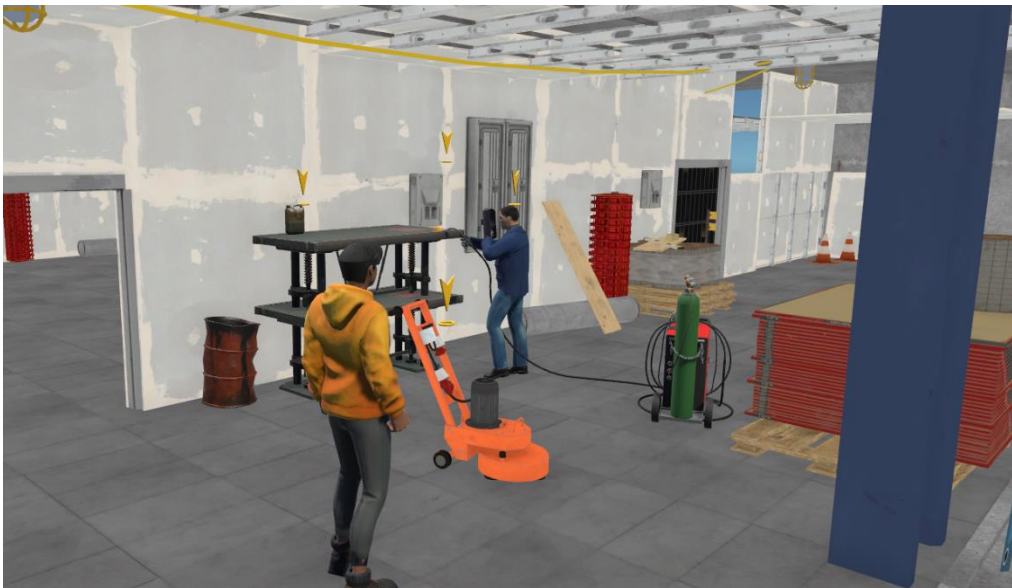


Figure 3-5. Hazardous Area of Welding Safety Training VR Environment

Scaffolding Safety Training Platform: This VR environment is specially designed for scaffolding safety training. Participants interact with various hazards to enhance their hazard recognition and critical thinking skills. By utilizing immersive technology and interactive elements, players are

encouraged to explore a virtual scaffolding site, identifying, and analyzing potential hazards that could impact worker safety. The VR environment highlights specific scaffolding-related hazards, such as missing guardrails, fall hazards, struck by material, and open hatch to promote engagement and learning, ensuring users are well-prepared to identify safety issues in scaffolding contexts as shown in Figure 3-6.



Figure 3-6. Hazardous Area of Scaffolding Safety Training VR Environment

In each of the three virtual training environments, an iSafeTrainer is available to assist participants based on their selected trade and levels of knowledge. iSafeTrainer can ask the questions that the students accurately identified all hazards and didn't miss any of them, it will ask them to completely identify and mitigate the hazards, but if they fail in identifying or

mitigating the hazards then iSafeTrainer can also suggest video-based training content relevant to the player specific requirements. These videos are integrated within the Unity-based VR environment and allow the worker to play and pause the video. This feature is intended to boost engagement and learning in the VR setting, as iSafeTrainer provides not just answers but also tailored educational content, as depicted in Figure 3-7.



(a)



(b)

Figure 3-7. (a) Conversational View Between iSafeTrainer and Player (b) Animated Video-based Learning Content for Training

3.1.2. LLM-based iSafeTrainer Avatar Creation

In the development of the proposed VR system, the primary objective was to incorporate an LLM-based virtual instructor as an Intelligent Safety Trainer (iSafeTrainer) to augment the training experience for students and workers. This incorporation aims to enhance the training through interactive and personalized communication with iSafeTrainer and delivery of training content. The design of the content adheres to the principles of constructivist learning theory, which underscores the importance of self-experiences and knowledge gain through environmental interaction, with the instructor [61].

The system utilized "Ready Me Player," a comprehensive library of customizable virtual characters, to develop the virtual instructor role. This virtual instructor is tailored to meet the specific communication needs of students and workers, thereby facilitating more effective acquisition of safety knowledge [66]. The virtual instructor is not only designed on the Ready Player Me platform but also animated within the Unity gaming environment. The animations, including idle and talk motions, are sourced from Mixamo, which offers a range of publicly accessible humanoid animations [67]. An animator component is integrated with iSafeTrainer, activating hand and leg movements during communication to mimic a realistic human like person. To further enhance realism, scripts for eye and head movements are attached, enabling it to blink and direct its gaze towards the students/worker (player) while engaging in dialogue and OVR lip synchronization of oculus has been

utilized to make the iSafeTrainer to open mouth for speaking [68]. The whole process of iSafeTrainer creation, animation with human like characteristics has been illustrated in Figure 3-8.

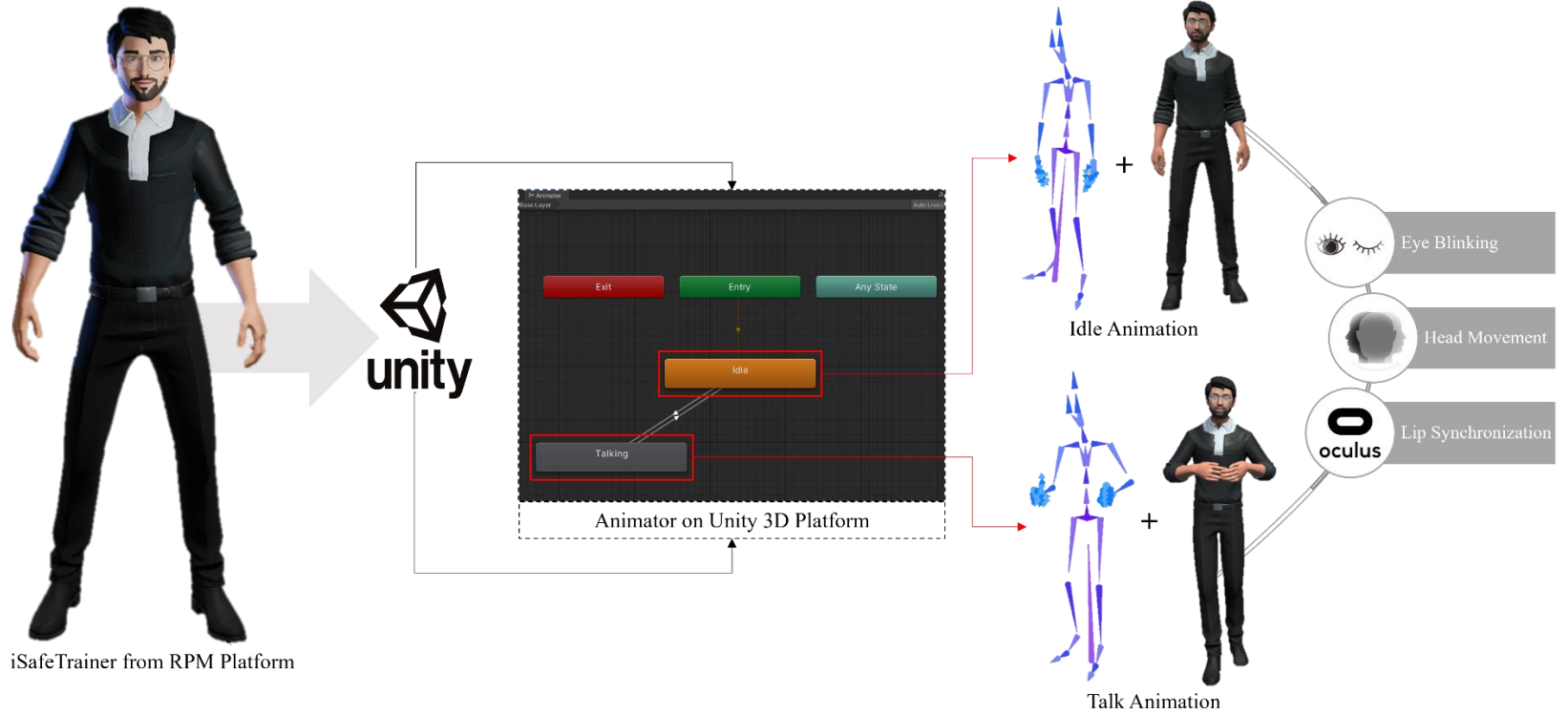


Figure 3-8. Animated iSafeTrainer Creation Procedure

3.2. Practical Implementations

Existing construction safety VR training programs mainly focus on delivering customized content through a series of pre-defined courses and pre-plotted VR tasks and generalized AI NPC for communication-based training for construction students and workers. However, offering personalized safety training emerges as a more proactive method, particularly in preparing individuals for the challenging realities of real-world construction environments [14]. This study embraces a communication-based approach, driven by LLM-based iSafeTrainer which is perfectly designed for the needs of individual, to provide personalized training. This approach is further enhanced with video-based demonstrations of training, allowing construction workers to watch and engage with content that is directly relevant to their expertise and experience in the field.

3.2.1. Prompt Engineering for iSafeTrainer

The iSafeTrainer is called intelligent based on its AI and prompt-based training to get to know about the user selection from the main menu through '*PlayerInfo*' script that which trade they have been selected and what is their level of knowledge for that trade either the students/workers (players) are a novice, beginner, or expert so that the communication can be initialized based on their selection. Through the prompts iSafeTrainer has been aware about each trade and the relevant hazards in the environment through the '*worldInfo*'

script such as “*GamStory*” and “*SimulatedEnvironment*” and whenever a student/worker asks the iSafeTrainer about any hazards it will provide to the point easy to understand answers to him, it has the information that how many and what kind of hazards are present in the user selected environment, so it will ask the student/worker that how many hazards he identified and to go back to identify the remaining hazards.

Prompt engineering-based training is a strategic process utilized to guide models like ChatGPT in understanding and generating relevant responses [69]. It involves adding specific prompts to extract desired outcomes from the model. Prompts serve as instructions for an LLM to enforce rules, automate processes, and ensure the generation of specific qualities and quantities of output [70]. By providing a well-designed prompt, the model can comprehend the intended task and deliver accurate and efficient responses. However, crafting an effective prompt is crucial. A well-designed prompt ensures that ChatGPT understands the task and generates contextually appropriate responses aligned with the specific needs and contexts of the construction students/workers [71]. Therefore, the proposed system incorporates prompt engineering techniques using the C# programming in Unity. These prompts not only include the player’s trade and their level of knowledge (based on what the player chooses from the main menu), but they also vividly describe the environment and story of the game which helps to immerse players in a fun and educational learning experience. Additionally, the prompts are intelligently tailored to

accommodate the limitations inherent to an AI entity, preventing any disclosure of its true nature and actively encouraging adherence to the predetermined identity established within the virtual realm of the game. Figure 3-9 shows how prompts based on C# scripting are aligned and integrated in Unity 3D game engine.

Moreover, the proposed VR system utilizes personalized instructions for the unique needs of trainees, by leveraging virtual reality (VR) simulations to create an engaging and realistic training environment. The iSafeTrainer based VR system leverages ChatGPT through OpenAI's official API, which has undergone training on a diverse internet corpus, enhancing its capacity to understand a broad range of user inputs, including those with spelling and grammatical variations [72]. However, when input lacks clarity or context, the resultant response might be inaccurate or nonsensical. To enhance the coherence of generated responses, rigorous validation of NLP models is needed. Although the comprehensive verification of NLP models extends beyond the scope of this thesis, the study employs prompt engineering just to address the needs of the workers for personalized training.

Furthermore, Whisper not only transcribes and translates voice input into English for the proposed VR system but it can perform same functionality in case of multilingual approach as used in previous study [19], [73] before passing it to the ChatGPT module. This multifaceted approach ensures a comprehensive exploration and integration of NLP capabilities within the

context of the research. But for now, the proposed study is not focusing on multilingual approach, but its purpose is the personalized training based on only trade and level of knowledge of the intended user.

Figure 3-9 provides a flow of the Voice-to-Text-to-Speech interaction process, outlining how voice input is converted to text, processed by ChatGPT using a chat prompt that includes instructions for role play, the game story, the simulated environment, and player details. The generated reply is logged and converted back to voice, making the system's final response. Additionally, if the worker asks for any kind of learning content which can be helpful for them to know more about that hazards the iSafeTrainer can recommend by informing them that the environment has a video-based training content for the users for further detailed training overview.

3.2.2. Player Information

Player information is managed by a C# script integrated with the iSafeTrainer system to enhance interaction specificity by identifying user requirements. The details selected by the user from the main menu are stored in the `PlayerInfo` script. This script dynamically updates its variables based on the trade and knowledge level selections of the user, which are then transmitted to the iSafeTrainer prompts which enables the system to tailor interactions and responses according to the specific user selections. This mechanism ensures that iSafeTrainer remains updated on the choices of the

user, allowing for more personalized and contextually relevant interactions within the training environment.

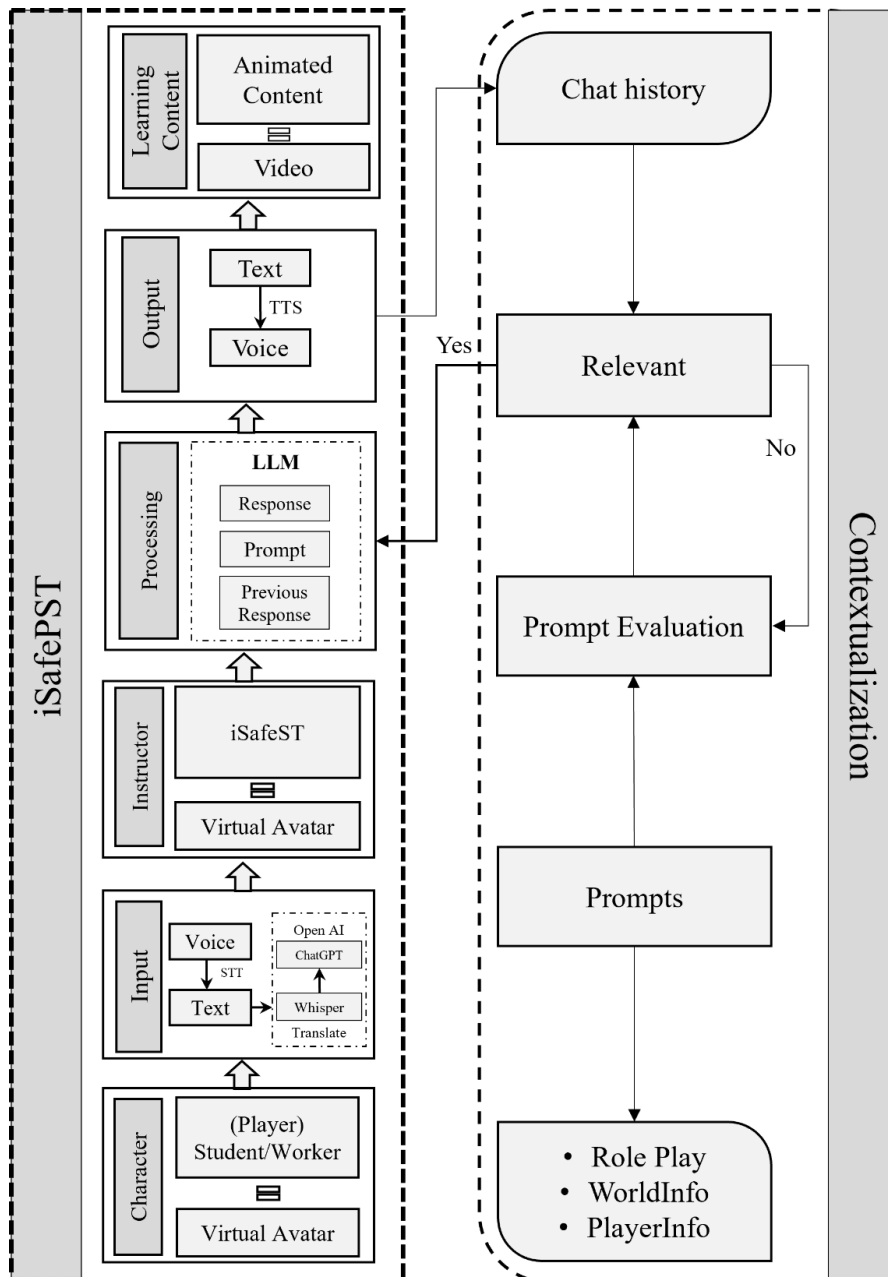


Figure 3-9. Voice-to-Text-to-Speech Interaction Process

3.2.3. AI Tools Integration for Personalized Safety Training

To address communication needs of students/workers and facilitate effective knowledge acquirement, an LLM-based iSafeTrainer was developed and integrated within the system which utilizes state-of-the-art AI tools, including Text-to-Speech (TTS), Whisper, and ChatGPT leveraging their advanced capabilities in natural language processing and speech synthesis. These key features play pivotal roles in enabling interactive and human-centered conversations throughout the learning process. This section provides a comprehensive overview of the knowledge of the iSafeTrainer, highlighting the incorporation of various tools that contribute to the natural and immersive conversational experience for the workers.

Among the integrated tools, TTS technology, which utilizes advanced machine learning algorithms to convert text into realistic spoken audio. This TTS script enhances user engagement and accessibility by providing dynamic audio outputs for text-based content. It is particularly beneficial for educational applications, accessibility aids for the visually impaired, and interactive voice systems. The script connects to OpenAI, processing textual input into audio streams that enhance multimodal interactions and improve user experience [74]. Next is whisper, a specialized AI tool developed by OpenAI, plays a crucial role in facilitating the transcription and translation of user interactions [75]. When a user submits a voice input through the microphone, whisper initiates a complex tokenization process, breaking down the audio into discrete units

called tokens. The whisper architecture, implemented as an encoder-decoder transformer, converts the input audio into a spectrogram, which is then passed into an encoder. A decoder trained within the system predicts the corresponding text caption, enriched with special tokens that guide the single model to perform various tasks, including language identification, multilingual speech transcription, and speech translation to English or other languages.

First, whisper predicts the language being spoken which is represented by a unique token for each language but in this study the authors are utilizing English language only so no need to go in further details. Moreover, when an audio segment does not contain any speech, the model is trained to predict a token indicating this absence. Following this, the subsequent token specifies the task, either transcription or translation, denoted by specific tokens [73]. The selection of whisper was driven by its impressive zero-shot performance across diverse datasets and demonstrating its robustness in handling varied communication scenarios. Its ability to minimize errors and consistently deliver accurate results in different contexts highlights its suitability for the system's transcription and translation functionalities.

Following the transcription and translation process, the generated text is then routed to ChatGPT which utilizes its advanced natural language processing capabilities to generate an appropriate answer or response. As ChatGPT has been trained on vast amounts of text data, its responses to a wide range of topics may be irrelevant during conversations. To address this, the

proposed VR system employs prompt engineering by incorporating contextualization techniques to refine and tailor responses, aligning them with the specific needs of participants. Within this framework, ‘contextualization’ refers to the practice of enhancing responses through crafted prompts that incorporate contextual information. These prompts include specific descriptions of the surrounding environment relevant to construction domain. The section 3.2.1 comprehensively explores this process, providing a detailed examination of how these tailored prompts are constructed and utilized. Subsequently, the refined text-based response undergoes processing for text-to-speech conversion.

To ensure clarity and reproducibility of the technological approach, a detailed breakdown of the integration process of the tools utilized in this research is provided by a sequence diagram in Figure 3-10. This figure represents the flow of data and interactions between these tools including APIs, configuration settings, preprocessing steps, and data formatting requirements. The integration of TTS, whisper, and ChatGPT in the VR system enables a seamless flow from voice generation to voice reception. The whisper framework ensures accurate transcription and translation of user interactions, while ChatGPT employs its LLMs’ capabilities to generate relevant responses. This comprehensive approach would facilitate effective communication, comprehension, and engagement between the instructor and trainee, creating a more inclusive and accessible training environment. Figure 3-10 depicts the

entire cycle, illustrating the flow from voice input to voice output, encompassing transcription and translation, and text-to-speech conversion.

3.2.4. Training Content Delivery

Under the framework of constructivist learning theory, the proposed approach enables users to engage interactively with the environment through the LLM-based iSafeTrainer, facilitated by prompt engineering for personalized communication. The iSafeTrainer is equipped to recommend video-based content strategically placed within the gaming scenes to enhance the training experience. These videos according to the expertise of the users are accessible with clear directional arrows for guidance to lead them to relevant training content. This method not only helps in learning about the hazards detailed in section 3.1.1 but also engages users with the VR game-based environment and make them able to solve the risks and hazards, illustrating safe and unsafe situations on a virtual construction site.

Moreover, a C# script controls the video playback with functionalities like play and pause, triggered by the user pressing '*Shift + P*' to play and '*Shift + R*' to pause. The iSafeTrainer is programmed to recognize specific queries for video content defined in the prompts, ensuring that relevant training materials in the Unity environment are readily accessible to the user. This interactive sequence and its functionalities provide a clear understanding of the operation and user interaction of the system as shown in Figure 3-11.

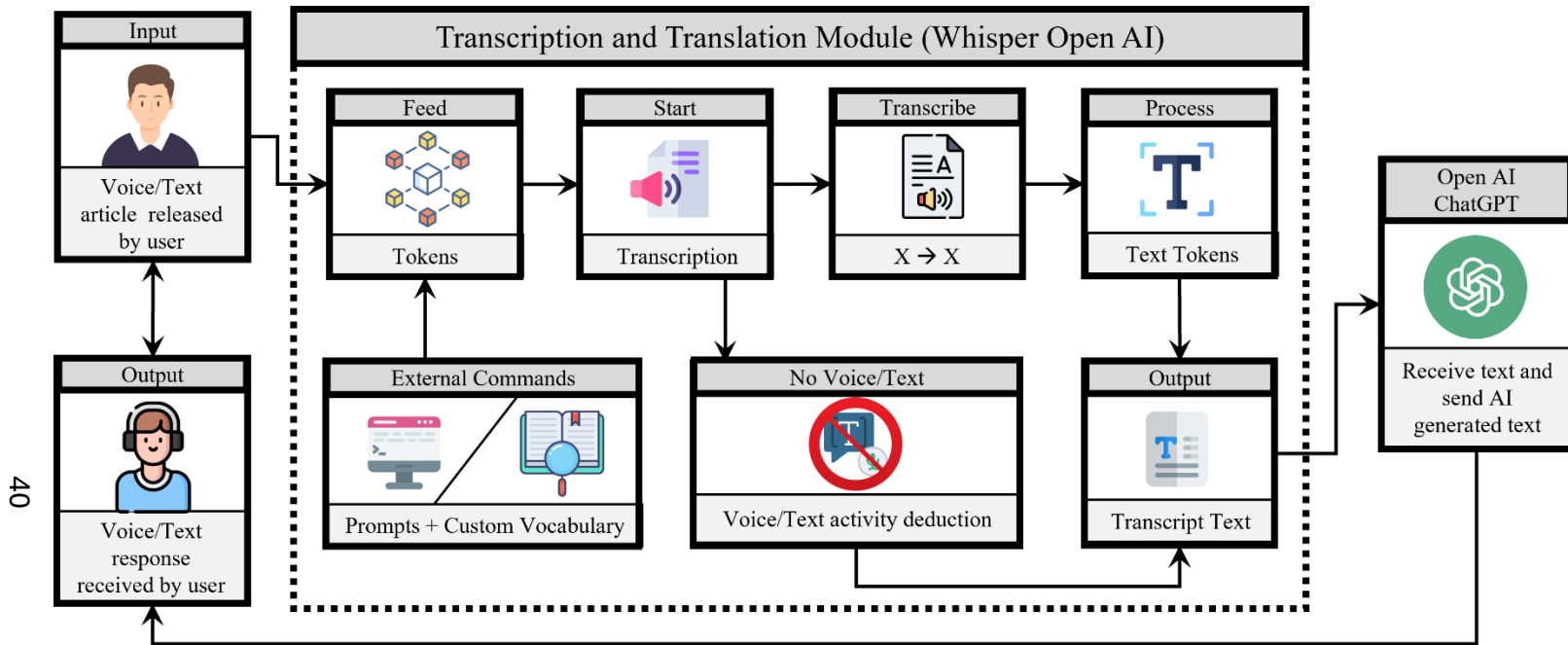


Figure 3-10. Process Flow of Voice Input to Voice Output for Transcription and Translation of Interaction Between Tools



(a)

```

void Update()
{
    if ((Input.GetKey(KeyCode.LeftShift) || Input.GetKey(KeyCode.RightShift)) && Input.GetKeyDown(KeyCode.P))
    {
        PlayVideo();
        UnityEngine.Debug.Log("Shift+P pressed. Playing video.");
    }

    if ((Input.GetKey(KeyCode.LeftShift) || Input.GetKey(KeyCode.RightShift)) && Input.GetKeyDown(KeyCode.R))
    {
        PauseVideo();
        UnityEngine.Debug.Log("Shift+R pressed. Pausing video.");
    }
}

private void PlayVideo()
{
    if (!videoPlayer.isPlaying)
    {
        videoPlayer.Play();
    }
}

private void PauseVideo()
{
    if (videoPlayer.isPlaying)
    {
        videoPlayer.Pause();
    }
}

```

(b)

Figure 3-11. (a) Learning Content in VR (b) C# Scripting to Control the Video Playback

4. Case Study and Experiments

4.1. Case Study

The proposed system approach enhances safety understanding and provides personalized learning through iSafeTrainer. Upon entering the VR environment, the user selects his trade such as Electrical, Scaffolding, or Welding and his level of knowledge either novice, beginner, or expert. For instance, if a user selects "Welding Safety Training" at the Beginner level, he starts the game by pressing the "Start" button to enter a virtual place populated with various hazards. These hazards are highlighted with popup arrows, designed to awaken curiosity, and encourage interaction with the environment. This setup helps beginners who may initially struggle to identify all hazards on their own.

During the session, a beginner might identify only three out of five hazards. He can then consult iSafeTrainer for feedback on his findings. The iSafeTrainer will inquire if all hazards have been identified, and upon hearing that only three were found by the student, it will inform him about the remaining hazards. The user is then encouraged to locate these newly discussed hazards, promoting an active learning process. This interactive feedback mechanism is critical for real-time evaluation, demonstrating sequential interaction and learning validation within the VR environment. Moreover, if the student fails to identify all hazards, iSafeTrainer can recommend video-

based content to foster a deeper understanding and enhance problem-solving abilities to identify and mitigate the hazards. This process is visually represented in the Error! Reference source not found..

Furthermore, Figure 4-2 shows how the player finds the hazard “struck by material” in Figure 4-2(a) which can cause a danger not only for the welder but also for common workers who passes by the welding area and in Figure 4-2(b) the player mitigated the hazard by placing it far away from the welding area, this process enhances the player’s critical thinking and problem-solving skills.

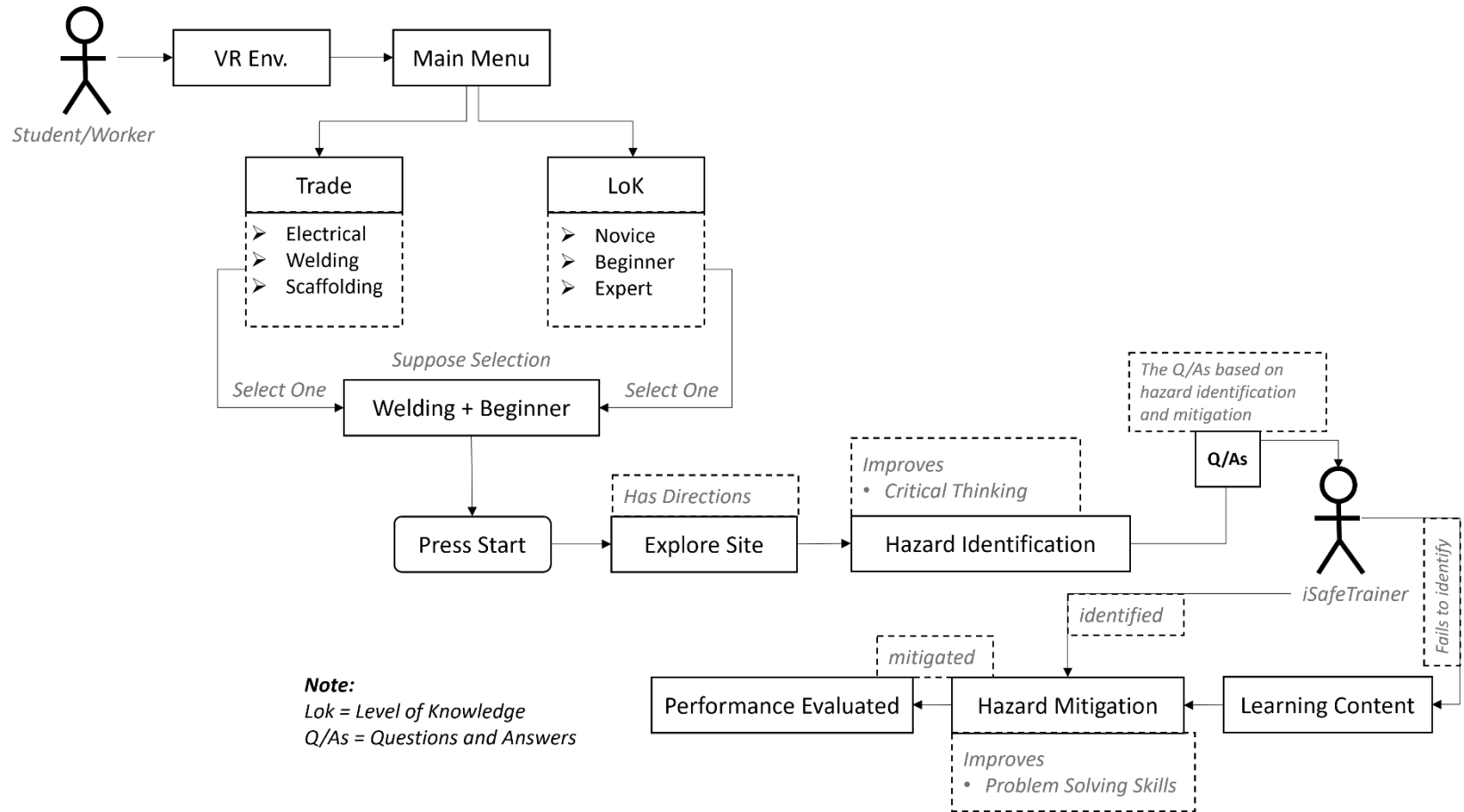


Figure 4-1. System Diagram and Case Study for Interaction with iSafeTrainer for Personalized Safety Training

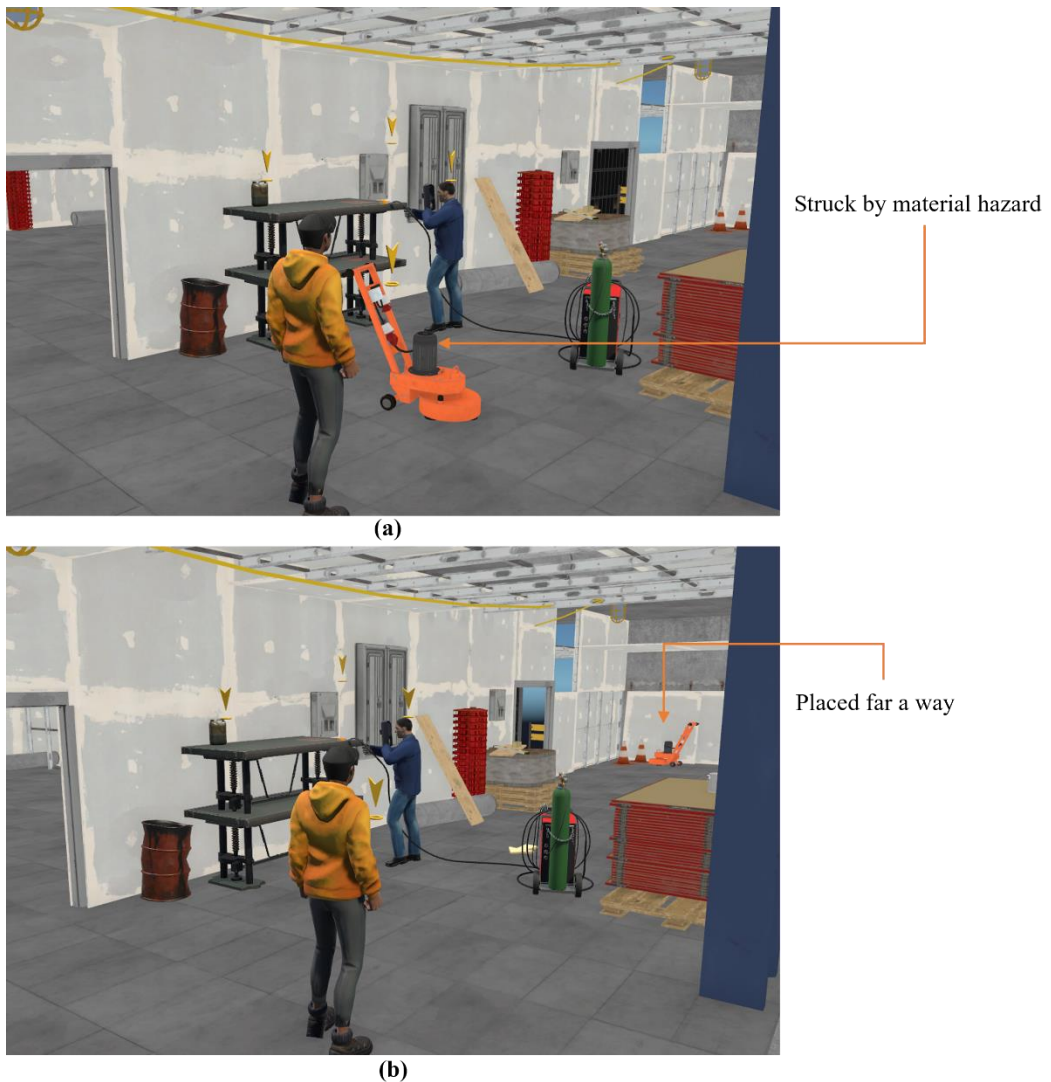


Figure 4-2. Identification of a struck by material hazard (b) Hazard mitigation

4.2. Experiments

After development of the iSafeTrainer based VR system, an experiment was conducted to evaluate its effectiveness for personalized safety training. Participants with different trades and levels of knowledge were asked to explore their respective virtual hazardous sites and then allowed them to

interact with the intelligent safety trainer for communication purposes. An evaluation of the system was conducted through an experiment using questionnaire-based surveys to assess participants' knowledge gain and satisfaction level after using the system. So, to comprehensively evaluate the effectiveness of the proposed VR-based system, a multi-faceted approach was adopted, incorporating both structured and unstructured survey questions. Structured questions, typically in the form of Likert scales were employed to gauge user satisfaction and general system usability. These structured questions allowed for a clear classification and quantification of responses, enabling statistical analysis and comparisons across participants [76].

A total of 31 participants, encompassing undergraduate, graduate students and post-doc candidates from Chung-Ang University, were meticulously recruited. To comprehensively represent the diverse worker population. The sample consisted of 23 males and 08 females. Table 4-1 summarizes the demographic information of the participants. The selection of participants for this study was meticulously approached to ensure comprehensive representation across critical demographic facets relevant to their trade and knowledge level. Further, the experiment consisted of two main components: a virtual reality (VR) simulation and a questionnaire-based survey.

Table 4-1: Background information of participants.

Demographic Factors	Number (n = 31)	Percentage (%)
Gender		
Male	23	74%
Female	08	25%
Age		
18 - 20	05	16%
25 - 30	19	61%
30 - 39	07	22%
Trade		
Electrical	14	45%
Welding	09	29%
Scaffolding	08	25%
Knowledge Level		
Novice	15	48%
Beginner	11	35%
Expert	05	16%

4.2.1. Virtual Reality Simulation with iSafeTrainer Interaction

Participants explored their respective virtual hazardous construction sites using the iSafeTrainer VR system, the same as discussed in section 4.1.1. These virtual environments were designed to replicate real-world construction

scenarios and potential safety hazards. During the simulation, participants were free to navigate and interact with the virtual environment, allowing them to identify and familiarize themselves with various safety hazards.

After or before exploring the virtual construction site, participants interacted with the LLM-based safety trainer incorporated into the VR system. This AI-powered trainer was designed to engage in natural language conversations, providing personalized safety guidance, explanations, and recommendations based on the participant's queries and the hazards encountered in the virtual environment. But key point to note here is that the virtual instructor is programmed to ask the player, if he explored the site before communication with it or not, if the user says no then the instructor will ask him to explore it first and then come to it with his queries. Participants communicated with the safety trainer using voice commands or text inputs, simulating a realistic training scenario. Figure 4-3 shows the participant communicating with iSafeTrainer in unity-based VR system for personalized safety training.

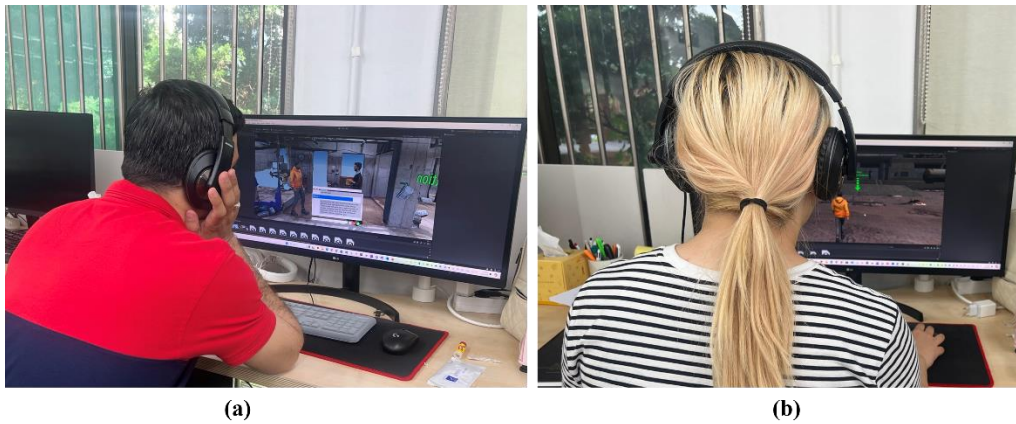


Figure 4-3. (a) Participant with expert level of knowledge having conversation with iSafeTrainer on electrical safety training platform (b) Participant with beginner level of knowledge is exploring the scaffolding safety training platform.

4.2.2. Questionnaire-based Survey

To assess the effectiveness of the iSafeTrainer system, a comprehensive questionnaire-based survey was conducted. The survey comprised a combination of structured and unstructured questions, designed to evaluate five key factors:

- User Satisfaction and Overall Usability
- User Experience and Interaction
- Training Effectiveness and Engagement
- System Guidance and Support
- Impact on Confidence and Preparedness

The structured questions utilized Likert scales and agreed/disagreed formats to gather quantitative data on various aspects, such as overall

satisfaction, ease of interaction, perceived realism, usefulness of training content, and likelihood of recommendation. Validated instruments, including the System Usability Scale (SUS) [77] and the User Experience Questionnaire (UEQ) [78], were incorporated to provide standardized measures of usability and user experience.

In addition to structured questions, unstructured questions were included to assess participants' critical thinking and problem-solving abilities after engaging with the VR system [79]. These questions allowed participants to express their thoughts, experiences, and insights freely, providing qualitative data on topics such as voice interaction experience, personalized training engagement, system guidance effectiveness, and impact on confidence in addressing safety hazards in real-world construction problems.

The combination of the VR simulation with LLM-based safety trainer interaction and the questionnaire-based survey enabled a comprehensive evaluation of the iSafeTrainer system's effectiveness in enhancing safety knowledge, providing personalized safety training, and fostering critical thinking and problem-solving skills among construction students and workers. After evaluating the knowledge gain and satisfaction level of participants, these surveys helped the author to understand the intended users need and wants, and the technical mistakes which have been addressed later after the questionnaire-based survey.

5. Evaluation Results

The novelty of the proposed system has been identified by comparing its qualitative attributes with recent studies in the literature, focusing on their adopted training methods and technologies used (Table 5-2). Further, to comprehensively evaluate the effectiveness of the proposed system in providing personalized safety training, a questionnaire-based survey was conducted, which was administered to a sample of 31 construction students who participated in the iSafeTrainer VR training program. The structured questions (1-8) utilized rating scales and agree/disagree formats to gather quantitative data on various aspects and the unstructured questions (9-15) allowed participants to express their thoughts, providing qualitative. Following is the detailed questionnaire in Table 5-1, and then a full summary of the questionnaire-based surveys with the mean results of each question in Table 5-3 along a thorough discussion of the experimental results has been presented.

Table 5-1. Full Questionnaire

Questionnaire-based Survey	
Structured Questions:	
1.	Rate your overall satisfaction with the iSafeTrainer system. (1 = Very Dissatisfied, 2 = Dissatisfied, 3 = Neutral, 4 = Satisfied, 5 = Very Satisfied).
2.	How easy was it to interact with the iSafeTrainer using voice commands? (1 = Very Difficult, 2 = Difficult, 4 = Normal, 4 = Easy, 5 = Very Easy)
3.	The iSafeTrainer provided personalized safety training tailored to my trade and knowledge level. (1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)

4. I found the virtual reality environment realistic and immersive.
(1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)
5. Rate the usefulness of the video-based training content recommended by the iSafeTrainer.
(1 = Not Useful at All, 2 = Not Useful, 3 = Normal, 5 = Useful, 5 = Extremely Useful)
6. How likely are you to recommend the iSafeTrainer system to other construction workers?
(1 = Not at All Likely, 2 = Not Likely, 3 = Neutral, 4 = Likely 5, = Extremely Likely)
7. The iSafeTrainer system improved my understanding of safety hazards and mitigation strategies in terms of personalized safety training.
(1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree)
8. Overall, how would you rate the usability of the iSafeTrainer system?
(1 = Very Poor, 2 = Poor, 3 = Normal, 4 = Good, 5 = Excellent)

Unstructured Questions:

9. Describe your experience interacting with iSafeTrainer. What worked well, and what could be improved?
10. Can you provide an example of a safety hazard you encountered in the virtual environment and how the iSafeTrainer guided you in mitigating it?
11. What did you find most engaging or beneficial about the personalized safety training provided by the iSafeTrainer?
12. If you could suggest any additional improvements to the iSafeTrainer system, what would they be?
13. How did you experience the iSafeTrainer system compared to traditional safety training methods?
14. Describe a situation where the iSafeTrainer's response or guidance was particularly helpful.
15. How did the iSafeTrainer system impact your confidence and preparedness in addressing safety hazards on construction sites?

Table 5-2. Comparison of different studies with the attributes of the proposed system

Year	Research Study	Training Method			Technology Adopted		Experimental Results		Reference
		One-Size	Personalized	Content Delivery	VR	Conversational-AI (LLMs)	Group	Survey	
2023	Li et. al.	x	o	x	o	x	o	x	[14]
2023	Wen et.al.	o	x	x	o	o	x	o	[38]
2024	Rane et. al.	o	x	x	x	o	x	o	[80]
2024	Hussain et. al.	o	x	o	o	o	o	x	[19]
2024	Proposed Study	x	o	o	o	o	o	o	

User Satisfaction and Overall Usability: The results from questions 1, 5, 7, and 8 indicated a good level of overall user satisfaction with the iSafeTrainer system. 81% of participants reported being "satisfied" or "very satisfied" with the system (Q1), 81% agreed on usefulness of provided training content, and 80% and 85% "agreed" or "strongly agreed" that the system improved their understanding of safety hazards and mitigation strategies (Q7 & Q8).

User Experience and Interaction: Analysis of questions 2, 9, and 12 revealed insights into the user experience and interaction with the system. Most participants (85%) found the voice interaction to be "easy" or "very easy" (Q2). However, the open-ended responses in Q9 and Q12 highlighted some areas for improvement and additional features including animations like, talk, walk, or head movement for realistic human nature of iSafeTrainer.

Training Effectiveness and Engagement: The survey results demonstrated the effectiveness and engagement of the personalized safety training provided by iSafeTrainer. 79% of participants "Neutral", "agreed" or "strongly agreed" that the system provided personalized training tailored to their trade and knowledge level (Q3), and 83% found the virtual environment realistic and immersive (Q4). Additionally, the open-ended responses in Q11 and Q13 revealed that participants found communication for real-time guidance and feedback to be particularly engaging and beneficial aspects of the personalized training. Compared to traditional safety training methods, participants

highlighted that trade and knowledge level-based training with AI-NPC is useful for them.

System Guidance and Support: 78% to 80% Participants provided examples of safety hazards encountered in the virtual environment and described how the iSafeTrainer guided them in mitigating these hazards (Q10). Common themes included how to place missing guardrails for scaffolding, which material is more dangerous in case of welding etc. Furthermore, in Q14, participants described situations where the system's guidance was particularly helpful, such as user asked to iSafeTrainer, I need some relevant training content to deal with overhead power lines, then iSafeTrainer suggested video-based training content from the VR environment to watch and recommended to the user to mitigate this hazard by removing or covering the overhead electrical wires. These responses suggest that the system provided effective guidance and support in real-world scenarios.

Impact on Confidence and Preparedness: The survey results indicated a positive impact of the iSafeTrainer system on users' confidence and preparedness in addressing safety hazards on construction sites. 75% of participants reported being “Neutral”, "likely" or "very likely" to recommend the system to other construction workers (Q6), suggesting a good level of confidence in the system's effectiveness.

Table 5-3: Overall questionnaire survey-based results

Questions	Mean	Results
1. Rate your overall satisfaction with the iSafeTrainer system on a scale of 1 to 5.	4.1	Satisfied Score illustrates that users felt at ease and satisfied when using the virtual screen for learning.
2. How easy was it to interact with the iSafeTrainer using voice commands?	4.3	Easy Score shows it was easy to interact with iSafeTrainer using voice commands but needs improvement
3. The iSafeTrainer provided personalized safety training tailored to my trade and knowledge level.	4.1	Agree iSafeTrainer provided most relevant and adequate answers.
4. I found the virtual reality environment realistic and immersive.	4.1	Agree The score describes those participants who mostly agreed on the realistic nature of VR environment.
5. Rate the usefulness of the video-based training content recommended by the iSafeTrainer.	4.0	Normal It seems that users can easily familiarize themselves with using the keyboard, mouse, and headphones to operate the system.
6. How likely are you to recommend the iSafeTrainer system to other construction workers?	3.7	Likely It seems that proposed system can be recommended to others.
7. The iSafeTrainer system improved my understanding of safety hazards and mitigation strategies in terms of personalized safety training.	4.0	Agree The personalized learning, communication and interaction of the system help users to pay attention and remember tailored safety knowledge.

8. How would you rate the usability of the iSafeTrainer system?	4.3	Good Seems like iSafeTrainer system is useful according to the personalized training requirements.
9. Describe your experience interacting with the iSafeTrainer. What worked well, and what could be improved?	4.0	Good Interact with intelligent safety trainer was helpful and most of the communication seems effective.
10. Can you provide an example of a safety hazard you encountered in the virtual environment and how the iSafeTrainer guided you in mitigating it?	4.0	Likely Most of the players provided examples of hazards they faced in the VR environment and tried to mitigate them.
11. What did you find most engaging or beneficial about the personalized safety training provided by the iSafeTrainer?	4.2	Useful Answers shows the tailored communication between iSafeTrainer, and player was more beneficial
12. If you could suggest any additional improvements to the iSafeTrainer system, what would they be?	4.0	Helpful Improvements related iSafeTrainer's animations like, talk, walk, or head movement was provided
13. How did you experience the iSafeTrainer system compared to traditional safety training methods?	4.0	Very good Very good scores highlight that users find the proposed more effective for personalized training as compared to traditional methods
14. Describe a situation where the iSafeTrainer's response or guidance was particularly helpful.	3.7	Clear A clear situation has been described in which the user got the effective guidance from iSafeTrainer.
15. How did the iSafeTrainer system impact your confidence and preparedness in addressing safety hazards on construction sites?	3.9	Satisfied Results shows that this system made the players enough capable to deal with real -world hazards effectively.

Additionally, the open-ended responses in Q15 revealed that the system realistic, collaborative, and personalized nature helped the novice and beginner players in terms of improving their confidence and preparedness for real-world jobsite situations.

In addition to the previously provided overall mean of questionnaire-based surveys and the results based on the five discussed factors, group surveys were also evaluated based on the participants' level of knowledge. The number of the participants corresponding to specific their knowledge level has been presented in Table 5-4 (here KF indicates Key Factor) with a graphical representation in Figure 5-1.

Table 5-4. Results of all five key factors corresponding to the level of knowledge of all 31 participants.

Key Factors	Novice	Beginner	Expert
KF1: User Satisfaction and Overall Usability	86.3%	82.3%	68.0%
KF2: User Experience and Interaction	87.6%	77.6%	73.3%
KF3: Training Effectiveness and Engagement	87.7%	80.5%	62.0%
KF4: System Guidance and Support	86.0%	76.4%	64.0%
KF5: Impact on Confidence and Preparedness	89.3%	69.1%	52.0%

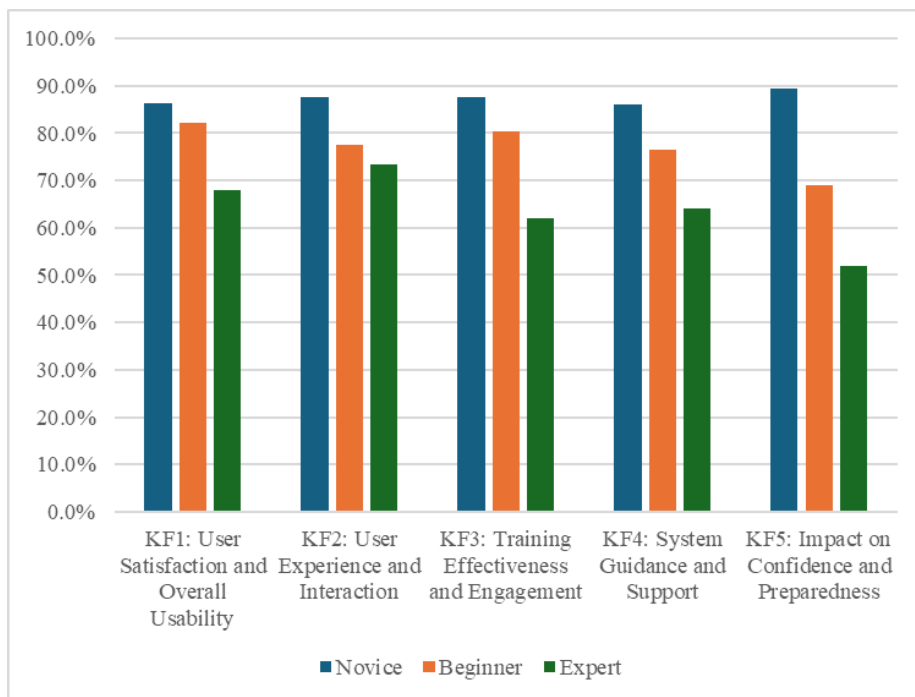


Figure 5-1. Graphical Representation of Table 5-4 Results

The bar chart in Figure 5-1 presents the evaluation of five key factors (KF) related to user satisfaction and training effectiveness across three user groups: Novice, Beginner, and Expert. Overall, Novices rated the system highest across all key factors, with scores consistently around or above 80%. Beginners provided slightly lower ratings, typically between 70% and 80%, indicating a positive yet less enthusiastic reception compared to Novices. Experts, however, showed the most critical stance, with ratings generally between 60% and 70%, particularly lower for KF3 (Training Effectiveness and Engagement) and KF5 (Impact on Confidence and Preparedness), which were around 60%. These results suggest that while the system is well-received, especially by less

experienced users, there is room for improvement to better meet the expectations of more experienced users.

6. Discussion

The proposed iSafeTrainer system introduces a novel approach to construction safety training by integrating advanced technologies such as virtual reality (VR) and conversational-AI (LLMs). This system addresses significant limitations present in literature by offering a more personalized and effective training solution. The LLMs (like Google Cloud, Microsoft Azure, and ChatGPT etc.) adopted training methods employed by Wen et al. (2023) and Rane et al. (2024) followed a one-size-fits-all approach [38], [80]. But the iSafeTrainer system distinguishes itself by adapting a tailored training approach to the specific trade and knowledge level of each worker. This personalized training method ensures that all users with different knowledge levels and trades receive appropriate instructions and training content, enhancing their learning outcomes. Wen et al. (2023) incorporated conversational AI but did not fully utilize this technology to personalize the learning experience [38]. The proposed system overcomes this limitation by leveraging LLM like OpenAI to provide personalized communication with iSafeTrainer by prompt engineering that meets the specific needs of each user.

The scope of the proposed study was comprehensive, involving a series of experiments designed to evaluate the effectiveness of the system and user satisfaction. The system integrates modern AI technologies such as Whisper for precise speech recognition and ChatGPT for interactive and personalized

dialogue. Whisper accurately transcribes and comprehends spoken instructions and inquiries, fostering smooth communication and inclusivity. ChatGPT functions as a virtual instructor, offering immediate feedback, resolving inquiries, and guiding trainees throughout the training process. This personalized approach boosts the effectiveness of knowledge acquisition, leading to an immersive and influential learning journey.

Participants in the study chose their desired safety training modules based on their trade and knowledge levels, allowing for a tailored training experience. Evaluation results indicated high degrees of satisfaction among users, particularly novices, who found the system's integration of LLM and immersive VR technology to be both engaging and effective. This aligns with Admi et al. (2023), which suggests that students perform better in VR-based training compared to traditional methods [81]. Users reported improved understanding of safety hazards and mitigation strategies and found the virtual environments realistic and immersive. The personalized video-based training content, tailored to their trades and knowledge levels, was well engaging.

The survey-based experiment results also identified areas for improvement based on participant feedback. Experienced users suggested the need for more advanced training scenarios that better align with their extensive industry knowledge and skills. This feedback underscores the importance of continually refining the system to meet the evolving needs of all user groups. Future

enhancements could include more sophisticated training modules that offer deeper, more complex simulations for expert professionals.

The proposed system represents a significant advancement in personalized safety training. By integrating AI with VR simulations, it enhances the learning process for both students and workers. The high levels of user satisfaction, positive ratings for system usability, and perceived improvements in safety knowledge demonstrate the effectiveness of the system. The advanced technologies employed encourage better communication, understanding, and interactivity, leading to a significant learning experience tailored to the unique requirements of users. Future research should focus on further optimizing the system to better serve diverse user groups across all experience levels.

7. Conclusion, Limitations and Future Work

7.1. Conclusion

Unskilled laborers often face higher risks of accidents on job sites due to standardized training methods that may not adequately address their individual needs. Addressing the safety needs of these workers during construction operations is a significant challenge, particularly considering their varying skill levels. The primary aim of this research is to develop a system that can address these challenges by providing personalized safety training sessions for workers.

This study introduces an innovative VR system, based on LLM, which is designed to cater to the specific safety training requirements of individual workers. Building upon previous work [19], the proposed system incorporates an intelligent safety trainer into the training process, utilizing advanced LLMs and AI tools. The system offers interactive, visual, and immersive training experiences within a synchronized virtual environment. The effectiveness of the proposed VR system for knowledge gain and satisfaction of workers and addressing their diverse training requirements is evaluated in the study through questionnaire-based surveys. Using this experimental approach, the system demonstrates significant improvements in knowledge levels of the participants, tailored to their personal information and expertise in their respective domains.

This research contributes to both the practical application of educational technology in the construction industry and the theoretical advancement of

pedagogical frameworks. It represents the systematic integration of VR and LLMs, such as ChatGPT, into a unified training system. This system is specifically designed for personalized training of construction workers. While VR and LLMs are existing technologies, the novelty of this research lies in their unique integration and application to address specific educational and training challenges faced by workers.

The proposed VR system distinguishes itself through its ability to deliver personalized educational experiences via an iSafeTrainer powered by prompt-engineered AI algorithms. The AI component, serving as the core feature of the iSafeTrainer system, enables personalized interactions by tailoring responses to individual trainees' attributes, such as their trade and level of knowledge. For example, iSafeTrainer not only modifies its responses based on the player's selections and the specific nature of their inquiries but also enhances the personalized learning experience by recommending video-based training content. This dual approach ensures that each worker or student receives personalized training and guidance, combining conversational interaction with supplementary visual materials for comprehensive understanding. By analyzing player queries, iSafeTrainer adeptly modulates the complexity of its explanations to ensure an optimal learning experience tailored to the unique needs of everyone.

Such personalization is crucial for addressing the diverse educational and construction backgrounds, as well as the varying learning levels or capacities

of workers and students, thereby facilitating a more inclusive and effective training process. Due to its adaptability and ease of implementation, the VR system can be seamlessly integrated into various safety programs. This adaptability allows trainers to conveniently customize the system to meet specific training needs, representing a significant step towards accessible and efficient educational technology implementation.

7.2. Limitations

While this study represents a significant advancement in addressing the personalized safety training needs of workers, it is crucial to acknowledge its inherent limitations. The efficacy of the proposed iSafeTrainer based VR system is heavily reliant on the capabilities of the underlying VR and AI technologies. Some challenges, including hardware performance, software stability, and the accuracy of natural language processing, have the potential to hinder the overall user experience and training outcomes.

While LLMs can be trained on extensive datasets, there may still be limitations in terms of their depth and breadth of knowledge in the specific domain of construction safety. Ensuring that the virtual instructor possesses comprehensive and up-to-date safety expertise across all relevant domains may necessitate extensive training and curation of the language model. Furthermore, Effective safety training often requires empathy, encouragement, and the ability to gauge the emotional state of the user. Integrating these soft skills into

the conversational abilities of iSafeTrainer may be challenging, as current LLMs still have limitations in emotional intelligence and social awareness.

7.3. Future Work

For future developments in the enhancement of the proposed system could involve the integration of LLM-based virtual instructor capable of demonstrating safety tasks to trainees [82]. These intelligent agents can simulate real-world safety scenarios on the request of the trainee, offering interactive and informative demonstrations to enhance trainees' comprehension and retention of safety protocols. This integration not only represents a significant step towards enhancing safety training initiatives in the construction industry, empowering trainees with practical knowledge and skills essential for on-site safety, but also opens new possibilities for the advancement of safety training methodologies, ultimately contributing to improved safety standards within the construction sector.

In this study, authors manually placed videos within the environment and provided prompts to the iSafeTrainer regarding the training content, allowing it to recommend training videos to players. While this method was effective, significant advancements could be achieved by enabling the LLM-based instructor to autonomously recommend video-based safety training content. By accessing a web database, the instructor could dynamically identify and recommend the most relevant and up to date training videos, enhancing the

customization and effectiveness of safety programs. Current advancements in AI and machine learning demonstrate the efficacy of AI in curating content based on user preferences and behavior [83], [84]. Moreover, an LLM-based instructor with autonomous recommendation capabilities would offer scalable solutions applicable to diverse industries and training programs, which aligns with the prevailing trend towards automated, AI-driven solutions in the field of occupational safety and health training [85]. This integration ensures training programs remain up to date with the latest safety standards and practices, tailored to each learner's needs, leading to more effective and engaging training experiences.

References

- [1] W. S. Alaloul, M. S. Liew, N. A. W. A. Zawawi, and I. B. Kennedy, “Industrial Revolution 4.0 in the construction industry: Challenges and opportunities for stakeholders,” *Ain Shams Eng. J.*, vol. 11, no. 1, pp. 225–230, Mar. 2020, doi: 10.1016/j.asej.2019.08.010.
- [2] A. Albert, I. Jeelani, and K. Han, “Developing hazard recognition skill among the next-generation of construction professionals,” *Constr. Manag. Econ.*, vol. 38, no. 11, pp. 1024–1039, Nov. 2020, doi: 10.1080/01446193.2020.1797133.
- [3] S. Hussain, W. Xuotong, and T. Hussain, “Impact of Skilled and Unskilled Labor on Project Performance Using Structural Equation Modeling Approach,” *SAGE Open*, vol. 10, no. 1, Jan. 2020, doi: 10.1177/2158244020914590/ASSET/IMAGES/LARGE/10.1177_2158244020914590-FIG2.JPEG.
- [4] M. J. Ding, F. Jie, S. Sisombat, and B. S. Bandlamudi, “Impact of the skill shortage on the construction supply chain performance in Australia,” *Civ. Eng. J.*, vol. 9, no. 2, p. 356, Feb. 2023, doi: 10.28991/CEJ-2023-09-02-08.
- [5] S. D. Boateng, D. A. Ankrah, and S. A. Manteaw, “Competence-based education: reflections on the context of teaching agriculture in Ghana’s pre-tertiary schools,” *Cogent Educ.*, vol. 10, no. 1, Dec. 2023, doi: 10.1080/2331186X.2023.2207793.
- [6] W.-D. Yu, K.-C. Wang, and H.-T. Wu, “Empirical Comparison of Learning Effectiveness of Immersive Virtual Reality–Based Safety Training for Novice and Experienced Construction Workers,” *J. Constr. Eng. Manag.*,

vol. 148, no. 9, p. 04022078, Sep. 2022, doi: 10.1061/(asce)co.1943-7862.0002337.

- [7] I. S. Abotaleb *et al.*, “A framework to integrate virtual reality into international standard safety trainings,” *Eng. Constr. Archit. Manag.*, vol. ahead-of-p, no. ahead-of-print, 2023, doi: 10.1108/ECAM-09-2023-0969.
- [8] M. S. Mohammed, J. Killingsworth, and S. Shah, “An Evaluation of Training Delivery Methods’ Effects on Construction Safety Training and Knowledge Retention - A Foundational Study,” *Int. J. Constr. Educ. Res.*, vol. 17, no. 1, pp. 18–36, Jan. 2021, doi: 10.1080/15578771.2019.1640319.
- [9] P. Adami *et al.*, “Effectiveness of VR-based training on improving construction workers’ knowledge, skills, and safety behavior in robotic teleoperation,” *Adv. Eng. Informatics*, vol. 50, p. 101431, Oct. 2021, doi: 10.1016/j.aei.2021.101431.
- [10] A. M. Alnajim, S. Habib, M. Islam, H. S. AlRawashdeh, and M. Wasim, “Exploring Cybersecurity Education and Training Techniques: A Comprehensive Review of Traditional, Virtual Reality, and Augmented Reality Approaches,” *Symmetry (Basel)*, vol. 15, no. 12, p. 2175, Dec. 2023, doi: 10.3390/sym15122175.
- [11] E. Rey-Becerra, L. H. Barrero, R. Ellegast, and A. Kluge, “The effectiveness of virtual safety training in work at heights: A literature review,” *Appl. Ergon.*, vol. 94, p. 103419, Jul. 2021, doi: 10.1016/j.apergo.2021.103419.
- [12] L. Albert and C. Routh, “Designing impactful construction safety training interventions,” *Safety*, vol. 7, no. 2, p. 42, May 2021, doi: 10.3390/safety7020042.

- [13] C. Wang, H. Zhu, C. Zhu, X. Zhang, E. Chen, and H. Xiong, “Personalized Employee Training Course Recommendation with Career Development Awareness,” *Web Conf. 2020 - Proc. World Wide Web Conf. WWW 2020*, pp. 1648–1659, Apr. 2020, doi: 10.1145/3366423.3380236.
- [14] W. Li, H. Huang, T. Solomon, B. Esmaeili, and L. F. Yu, “Synthesizing Personalized Construction Safety Training Scenarios for VR Training,” *IEEE Trans. Vis. Comput. Graph.*, vol. 28, no. 5, pp. 1993–2002, May 2022, doi: 10.1109/TVCG.2022.3150510.
- [15] P. Duan, J. Zhou, and W. Fan, “Safety tag generation and training material recommendation for construction workers: a persona-based approach,” *Eng. Constr. Archit. Manag.*, vol. 31, no. 1, pp. 115–135, Jan. 2024, doi: 10.1108/ECAM-12-2021-1143.
- [16] M. Wolf, J. Teizer, B. Wolf, S. Bükürü, and A. Solberg, “Investigating hazard recognition in augmented virtuality for personalized feedback in construction safety education and training,” *Adv. Eng. Informatics*, vol. 51, p. 101469, Jan. 2022, doi: 10.1016/j.aei.2021.101469.
- [17] N. Tang, H. Hu, F. Xu, and F. Zhu, “Personalized safety instruction system for construction site based on internet technology,” *Saf. Sci.*, vol. 116, pp. 161–169, Jul. 2019, doi: 10.1016/j.ssci.2019.03.001.
- [18] A. Rahouti, R. Lovreglio, S. Datoussaïd, and T. Descamps, “Prototyping and Validating a Non-immersive Virtual Reality Serious Game for Healthcare Fire Safety Training,” *Fire Technol.*, vol. 57, no. 6, pp. 3041–3078, Nov. 2021, doi: 10.1007/s10694-021-01098-x.
- [19] R. Hussain *et al.*, “Conversational AI-based VR system to improve

- construction safety training of migrant workers,” *Autom. Constr.*, vol. 160, p. 105315, Apr. 2024, doi: 10.1016/j.autcon.2024.105315.
- [20] H. M. Tusher, S. Mallam, and S. Nazir, “A Systematic Review of Virtual Reality Features for Skill Training,” *Technol. Knowl. Learn.*, pp. 1–36, Jan. 2024, doi: 10.1007/s10758-023-09713-2.
- [21] L. Villa, D. Carneros-Prado, A. Sánchez-Miguel, C. C. Dobrescu, and R. Hervás, “Conversational Agent Development Through Large Language Models: Approach with GPT,” *Lect. Notes Networks Syst.*, vol. 835 LNNS, pp. 286–297, 2023, doi: 10.1007/978-3-031-48306-6_29/COVER.
- [22] Y. Chang *et al.*, “A Survey on Evaluation of Large Language Models,” *ACM Trans. Intell. Syst. Technol.*, vol. 15, no. 3, pp. 1–45, Jun. 2024, doi: 10.1145/3641289.
- [23] I. Kamardeen and A. Hasan, “Occupational Health and Safety Challenges for Sustaining Construction Apprentice Programs,” *J. Manag. Eng.*, vol. 38, no. 5, p. 04022042, Sep. 2022, doi: 10.1061/(asce)me.1943-5479.0001059.
- [24] M. J. Ritchie, L. E. Parker, and J. A. E. Kirchner, “From novice to expert: a qualitative study of implementation facilitation skills,” *Implement. Sci. Commun.*, vol. 1, no. 1, pp. 1–12, Dec. 2020, doi: 10.1186/s43058-020-00006-8.
- [25] D. Alexandrovsky *et al.*, “Examining Design Choices of Questionnaires in VR User Studies,” *Conf. Hum. Factors Comput. Syst. - Proc.*, Apr. 2020, doi: 10.1145/3313831.3376260.
- [26] A. Ojha, J. Seagers, S. Shayesteh, and M. Habibnezhad, “Construction Safety Training Methods and their Evaluation Approaches: A Systematic Literature

Review,” 2020, Accessed: Apr. 02, 2024. [Online]. Available:

<https://www.researchgate.net/publication/346856894>

- [27] A. D. Rafindadi, N. Shafiq, I. Othman, and M. Mikić, “Mechanism Models of the Conventional and Advanced Methods of Construction Safety Training. Is the Traditional Method of Safety Training Sufficient?,” *Int. J. Environ. Res. Public Health*, vol. 20, no. 2, Jan. 2023, doi: 10.3390/IJERPH20021466.
- [28] S. Zhang, K. Sulankivi, M. Kiviniemi, I. Romo, C. M. Eastman, and J. Teizer, “BIM-based fall hazard identification and prevention in construction safety planning,” *Saf. Sci.*, vol. 72, pp. 31–45, Feb. 2015, doi: 10.1016/J.SSCI.2014.08.001.
- [29] P. Bęś and P. Strzałkowski, “Analysis of the Effectiveness of Safety Training Methods,” *Sustain. 2024, Vol. 16, Page 2732*, vol. 16, no. 7, p. 2732, Mar. 2024, doi: 10.3390/SU16072732.
- [30] A. D. Rafindadi, N. Shafiq, I. Othman, and M. Mikić, “Mechanism Models of the Conventional and Advanced Methods of Construction Safety Training. Is the Traditional Method of Safety Training Sufficient?,” *Int. J. Environ. Res. Public Health*, vol. 20, no. 2, p. 1466, Jan. 2023, doi: 10.3390/ijerph20021466.
- [31] T. Åsgård and P. Nygaard, “From theoretical to practical competence on health and safety,” *Emerald Reach Proc. Ser.*, vol. 2, pp. 473–479, 2019, doi: 10.1108/S2516-285320190000002046.
- [32] X. Liu and X. Li, “Exploring the Formation Mechanism of Unsafe Construction Behavior and Testing Efficient Occupational Health and Safety (OHS) Programs,” *Int. J. Environ. Res. Public Health*, vol. 19, no. 4, 2022,

doi: 10.3390/ijerph19042090.

- [33] A. D. Kaplan, J. Cruik, M. Endsley, S. M. Beers, B. D. Sawyer, and P. A. Hancock, "The Effects of Virtual Reality, Augmented Reality, and Mixed Reality as Training Enhancement Methods: A Meta-Analysis," *Hum. Factors*, vol. 63, no. 4, pp. 706–726, Jun. 2021, doi: 10.1177/0018720820904229.
- [34] R. Hussain, S. F. A. Zaidi, A. Pedro, H. Lee, and C. Park, "Exploring construction workers' attention and awareness in diverse virtual hazard scenarios to prevent struck-by accidents," *Saf. Sci.*, vol. 175, Jul. 2024, doi: 10.1016/j.ssci.2024.106526.
- [35] M. Akinlolu and T. C. Haupt, "Effectiveness of BIM-based visualization technologies for construction site health and safety management: A Meta-synthesis Approach," *IOP Conf. Ser. Mater. Sci. Eng.*, vol. 1107, no. 1, p. 012176, Apr. 2021, doi: 10.1088/1757-899x/1107/1/012176.
- [36] M. Afzal, M. T. Shafiq, and H. Al Jassmi, "Improving construction safety with virtual-design construction technologies - A review," *J. Inf. Technol. Constr.*, vol. 26, pp. 319–340, 2021, doi: 10.36680/j.itcon.2021.018.
- [37] Y. Zhu and N. Li, "Virtual and augmented reality technologies for emergency management in the built environments: A state-of-the-art review," *J. Saf. Sci. Resil.*, vol. 2, no. 1, pp. 1–10, Mar. 2021, doi: 10.1016/j.jnlssr.2020.11.004.
- [38] J. Wen and M. Gheisari, "iVisit-Communicate for AEC Education: Using Virtual Humans to Practice Communication Skills in 360-Degree Virtual Field Trips," *J. Comput. Civ. Eng.*, vol. 37, no. 3, May 2023, doi: 10.1061/jccee5.cpeng-5165.

- [39] M. Jelonek *et al.*, “Evaluating Virtual Reality Simulations for Construction Safety Training A User Study Exploring Learning Effects, Usability and User Experience,” *I-Com*, vol. 21, no. 2, pp. 269–281, Aug. 2022, doi: 10.1515/icom-2022-0006.
- [40] S. Xu, Q. Q. Ni, M. Zhang, and M. Li, “A personalized safety training system for construction workers,” *Int. Conf. Smart Infrastruct. Constr. 2019, ICSIC 2019 Driv. Data-Informed Decis.*, pp. 321–326, 2019, doi: 10.1680/icsic.64669.321.
- [41] K. Lee, Y. Shinde, S. Hasanzadeh, and B. Esmacili, “Toward Personalized Safety Training: Automating the Classification of Construction Workers’ Cognitive Failures,” *Proc. Int. Symp. Autom. Robot. Constr.*, vol. 2022-July, pp. 268–275, 2022, doi: 10.22260/isarc2022/0038.
- [42] A. Anand, *Exploring the Applications and Limitations of Large Language Models: A Focus on ChatGPT in Virtual NPC Interactions*. 2023. Accessed: Apr. 02, 2024. [Online]. Available: <https://search.proquest.com/openview/97442be4967a379d622ab9ee53fd1557/1?pq-origsite=gscholar&cbl=18750&diss=y>
- [43] Y. Ge, Y. Ren, W. Hua, S. Xu, J. Tan, and Y. Zhang, “LLM as OS, Agents as Apps: Envisioning AIOS, Agents and the AIOS-Agent Ecosystem,” Dec. 2023, doi: 10.48550/arXiv.2312.03815.
- [44] D. Chen, H. Wang, Y. Huo, Y. Li, and H. Zhang, “GameGPT: Multi-agent Collaborative Framework for Game Development,” Oct. 2023, doi: <https://doi.org/10.48550/arXiv.2310.08067>.
- [45] M. Klissarov *et al.*, “Motif: Intrinsic Motivation from Artificial Intelligence

- Feedback,” Sep. 2023, Accessed: Apr. 09, 2024. [Online]. Available:
<http://arxiv.org/abs/2310.00166>
- [46] T. Händler, “Balancing Autonomy and Alignment: A Multi-Dimensional Taxonomy for Autonomous LLM-powered Multi-Agent Architectures,” Oct. 2023, Accessed: Apr. 09, 2024. [Online]. Available:
<http://arxiv.org/abs/2310.03659>
- [47] S. Hu *et al.*, “A Survey on Large Language Model-Based Game Agents,” Apr. 2024, doi: <https://doi.org/10.48550/arXiv.2310.00166>.
- [48] X. Xue *et al.*, “Computational Experiments Meet Large Language Model Based Agents: A Survey and Perspective,” *arXiv Prepr.*, pp. 1–24, Feb. 2024, Accessed: Apr. 09, 2024. [Online]. Available:
<https://arxiv.org/abs/2402.00262v1>
- [49] F. Li, X. Wang, B. Li, Y. Wu, Y. Wang, and X. Yi, “A Study on Training and Developing Large Language Models for Behavior Tree Generation,” Jan. 2024, Accessed: Apr. 09, 2024. [Online]. Available:
<http://arxiv.org/abs/2401.08089>
- [50] A. AUXTERO, “GAME ENVIRONMENT DESIGN CREATOR USING ARTIFICIAL INTELLIGENCE PROCEDURAL GENERATION,” 2023, Accessed: Apr. 02, 2024. [Online]. Available:
https://run.unl.pt/bitstream/10362/164247/1/Auxtero_2023.pdf
- [51] E. Bozkir *et al.*, “Embedding Large Language Models into Extended Reality: Opportunities and Challenges for Inclusion, Engagement, and Privacy,” *arxiv.org*, 2024, Accessed: Apr. 02, 2024. [Online]. Available:
<http://arxiv.org/pdf/2402.03907v1>

- [52] P. Sweetser, “Large Language Models and Video Games: A Preliminary Scoping Review,” Mar. 2024, Accessed: Apr. 02, 2024. [Online]. Available: <http://arxiv.org/abs/2403.02613>
- [53] I. Jeelani, K. Han, and A. Albert, “Development of virtual reality and stereo-panoramic environments for construction safety training,” *Eng. Constr. Archit. Manag.*, vol. 27, no. 8, pp. 1853–1876, Oct. 2020, doi: 10.1108/ECAM-07-2019-0391.
- [54] Z. Xu and N. Zheng, “Incorporating virtual reality technology in safety training solution for construction site of urban cities,” *Sustain.*, vol. 13, no. 1, pp. 1–19, Dec. 2021, doi: 10.3390/su13010243.
- [55] O. G. Dela Cruz and J. S. Dajac, “Virtual Reality (VR): A Review on its Application in Construction Safety,” *Turkish J. Comput. Math. Educ.*, vol. 12, no. 11, pp. 3379–3393, May 2021, doi: 10.17762/TURCOMAT.V12I11.6383.
- [56] R. Eiris, M. Gheisari, and B. Esmacili, “Desktop-based safety training using 360-degree panorama and static virtual reality techniques: A comparative experimental study,” *Autom. Constr.*, vol. 109, p. 102969, Jan. 2020, doi: 10.1016/j.autcon.2019.102969.
- [57] S. Joshi *et al.*, “Implementing Virtual Reality technology for safety training in the precast/ prestressed concrete industry,” *Appl. Ergon.*, vol. 90, p. 103286, Jan. 2021, doi: 10.1016/J.APERGO.2020.103286.
- [58] T. Kaarlela, S. Pieska, and T. Pitkaaho, “Digital twin and virtual reality for safety training,” *11th IEEE Int. Conf. Cogn. Infocommunications, CogInfoCom 2020 - Proc.*, pp. 115–120, Sep. 2020, doi:

10.1109/COGINFOCOM50765.2020.9237812.

- [59] A. Anand, “Exploring the Applications and Limitations of Large Language Models: A Focus on ChatGPT in Virtual NPC Interactions,” 2023.
- [60] J. Zajda, “Constructivist Learning Theory and Creating Effective Learning Environments,” pp. 35–50, 2021, doi: 10.1007/978-3-030-71575-5_3.
- [61] S. Chuang, “The Applications of Constructivist Learning Theory and Social Learning Theory on Adult Continuous Development,” *Perform. Improv.*, vol. 60, no. 3, pp. 6–14, Mar. 2021, doi: 10.1002/pfi.21963.
- [62] P. Bęś and P. Strzałkowski, “Analysis of the Effectiveness of Safety Training Methods,” *Sustain. 2024, Vol. 16, Page 2732*, vol. 16, no. 7, p. 2732, Mar. 2024, doi: 10.3390/SU16072732.
- [63] W. Li, X. Liu, Q. Zhang, B. Zhou, B. W.-A. Sciences, and undefined 2023, “VR-Enhanced Cognitive Learning: Method, Framework, and Application,” *mdpi.com* W Li, X Liu, Q Zhang, B Zhou, B Wang *Applied Sci.* 2023 • *mdpi.com*, Accessed: Apr. 16, 2024. [Online]. Available: <https://www.mdpi.com/2076-3417/13/8/4756>
- [64] “Sketchfab - Platform for 3D ZVirtual Objects.” <https://sketchfab.com/feed> (accessed Mar. 20, 2024).
- [65] B. Kerbl, G. Kopanas, T. Leimkuehler, and G. Drettakis, “3D Gaussian Splatting for Real-Time Radiance Field Rendering,” *ACM Trans. Graph.*, vol. 42, no. 4, 2023, doi: 10.1145/3592433.
- [66] “Ready Player Me, Cross-Game Avatar Platform for the Metaverse.” <https://readyplayer.me/> (accessed Mar. 06, 2024).
- [67] “Mixamo - animation provider.” <https://www.mixamo.com/#/> (accessed Mar.

25, 2024).

- [68] “Oculus Lipsync Unity.”
<https://developer.oculus.com/downloads/package/oculus-lipsync-unity/>
(accessed Apr. 26, 2024).
- [69] J. White *et al.*, “A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT,” Feb. 2023, Accessed: Mar. 26, 2024. [Online]. Available: <http://arxiv.org/abs/2302.11382>
- [70] P. Liu, W. Yuan, J. Fu, Z. Jiang, H. Hayashi, and G. Neubig, “Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing,” *ACM Comput. Surv.*, vol. 55, no. 9, Jan. 2023, doi: 10.1145/3560815.
- [71] L. Reynolds and K. McDonell, “Prompt Programming for Large Language Models: Beyond the Few-Shot Paradigm,” *Conf. Hum. Factors Comput. Syst. - Proc.*, May 2021, doi: 10.1145/3411763.3451760.
- [72] A. Neelakantan *et al.*, “Text and Code Embeddings by Contrastive Pre-Training,” Jan. 2022, Accessed: Mar. 26, 2024. [Online]. Available: <http://arxiv.org/abs/2201.10005>
- [73] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust Speech Recognition via Large-Scale Weak Supervision,” *Proceedings of Machine Learning Research*, vol. 202. PMLR, pp. 28492–28518, Jul. 03, 2023. Accessed: Mar. 26, 2024. [Online]. Available: <https://proceedings.mlr.press/v202/radford23a.html>
- [74] IBM, “Text to Speech | FPT.AI.” <https://cloud.ibm.com/catalog/services/text-to-speech#about%0Ahttps://fpt.ai/tts> (accessed Apr. 26, 2024).

- [75] “Introducing Whisper - OpenAI.” <https://openai.com/research/whisper> (accessed Mar. 26, 2024).
- [76] L. Xiangming, K. Wang, Y. Wang, J. He, and J. Zhang, “Assessing User Experiences in Virtual Reality Learning Environments,” *Asia-Pacific Educ. Res.*, vol. 1, pp. 1–12, Jan. 2024, doi: 10.1007/S40299-023-00785-3/TABLES/1.
- [77] G. Gronier and A. Baudet, “Psychometric Evaluation of the F-SUS: Creation and Validation of the French Version of the System Usability Scale,” *Int. J. Human–Computer Interact.*, vol. 37, no. 16, pp. 1571–1582, Oct. 2021, doi: 10.1080/10447318.2021.1898828.
- [78] M. Schrepp, J. Thomaschewski, and A. Hinderks, “Construction of a Benchmark for the User Experience Questionnaire (UEQ),” *Int. J. Interact. Multimed. Artif. Intell.*, vol. 4, no. Regular Issue, pp. 40–44, 2017, doi: 10.9781/IJIMAI.2017.445.
- [79] Z. Merchant, E. T. Goetz, L. Cifuentes, W. Keeney-Kennicutt, and T. J. Davis, “Effectiveness of virtual reality-based instruction on students’ learning outcomes in K-12 and higher education: A meta-analysis,” *Comput. Educ.*, vol. 70, pp. 29–40, Jan. 2014, doi: 10.1016/J.COMPEDU.2013.07.033.
- [80] N. Rane, S. Choudhary, and J. Rane, “A New Era of Automation in the Construction Industry: Implementing Leading-Edge Generative Artificial Intelligence, such as ChatGPT or Bard,” *SSRN Electron. J.*, Jan. 2024, doi: 10.2139/SSRN.4681676.
- [81] P. Adami *et al.*, “Participants matter: Effectiveness of VR-based training on

the knowledge, trust in the robot, and self-efficacy of construction workers and university students,” *Adv. Eng. Informatics*, vol. 55, p. 101837, Jan. 2023, doi: 10.1016/J.AEI.2022.101837.

- [82] J. Varas *et al.*, “Innovations in surgical training: exploring the role of artificial intelligence and large language models (LLM),” *Rev. Col. Bras. Cir.*, vol. 50, p. e20233605, Aug. 2023, doi: 10.1590/0100-6991E-20233605-EN.
- [83] B. Kirdemir, J. Kready, E. Mead, M. N. Hussain, and N. Agarwal, “Examining Video Recommendation Bias on YouTube,” *Commun. Comput. Inf. Sci.*, vol. 1418, pp. 106–116, 2021, doi: 10.1007/978-3-030-78818-6_10/FIGURES/3.
- [84] F. Gaw, “Algorithmic logics and the construction of cultural taste of the Netflix Recommender System,” *Media, Cult. Soc.*, vol. 44, no. 4, pp. 706–725, May 2022, doi: 10.1177/01634437211053767/ASSET/IMAGES/LARGE/10.1177_01634437211053767-FIG3.JPEG.
- [85] I. A. Shah and S. D. Mishra, “Artificial intelligence in advancing occupational health and safety: an encapsulation of developments,” *J. Occup. Health*, vol. 66, no. 1, p. 17, Jan. 2024, doi: 10.1093/JOCCUH/UIAD017.

Appendix I

Table A-1. Number of Participants corresponding to their trade and level of knowledge.

Participant No.	Electrical	Welding	Scaffolding	Novice	Beginner	Expert
1	0			0		
2		0			0	
3	0			0		
4		0			0	
5	0			0		
6			0	0		
7			0			0
8	0			0		
9		0			0	
10	0			0		
11		0				0
12	0			0		
13			0		0	
14	0				0	
15			0	0		
16	0			0		
17		0				0
18	0				0	
19		0			0	
20			0	0		
21	0					0
22			0	0		
23		0			0	
24	0				0	
25			0		0	
26	0					0
27		0		0		
28	0				0	
29	0			0		
30			0	0		
31		0		0		

Table A-2. Results of each survey question correspond to key factors.

No.	KF1: User Satisfaction and Overall Usability				KF2: User Experience and Interaction			KF3: Training Effectiveness and Engagement				KF4: System Guidance and Support		KF5: Impact on Confidence and Preparedness		Results		
	Q#1	Q#5	Q#7	Q#8	Q#2	Q#9	Q#12	Q#3	Q#4	Q#11	Q#13	Q#10	Q#14	Q#6	Q#15	Score	Total Score	Percentage
1	4	3	4	5	3	5	4	3	5	3	4	3	4	4	5	59	75	78.67%
2	3	3	5	5	3	4	3	4	5	3	3	3	5	4	4	57	75	76.00%
3	5	4	4	5	4	5	3	4	5	4	5	5	3	5	5	66	75	88.00%
4	4	5	5	5	5	4	2	5	4	5	2	5	3	3	5	62	75	82.67%
5	5	5	3	5	5	5	5	4	5	5	5	5	5	3	5	70	75	93.33%
6	4	5	5	4	5	4	5	5	4	5	4	5	4	5	4	68	75	90.67%
7	4	4	4	2	4	3	4	2	2	4	3	3	4	3	3	49	75	65.33%
8	5	5	4	5	5	4	5	3	5	5	3	5	3	5	3	65	75	86.67%
9	4	4	4	5	4	5	3	5	5	4	3	4	3	3	3	59	75	78.67%
10	5	5	4	4	5	4	4	4	5	5	5	5	5	5	5	70	75	93.33%
11	2	2	3	4	2	4	5	3	3	2	4	2	4	3	3	46	75	61.33%
12	4	4	3	5	4	5	4	5	5	4	5	3	5	4	5	65	75	86.67%
13	5	5	3	4	5	3	3	3	3	3	3	5	4	4	4	57	75	76.00%
14	4	4	2	5	4	3	5	5	5	4	4	4	4	2	3	58	75	77.33%
15	4	5	5	4	5	4	4	4	4	5	5	5	5	3	4	66	75	88.00%
16	4	3	5	4	4	4	5	4	5	5	5	4	5	5	3	65	75	86.67%
17	3	3	4	3	4	4	4	3	2	4	2	4	2	2	4	48	75	64.00%
18	3	4	4	3	5	4	3	4	3	5	4	3	4	3	4	56	75	74.67%
19	3	4	5	3	5	4	4	2	5	3	3	3	3	3	4	54	75	72.00%
20	3	4	4	5	5	5	5	3	4	3	5	4	5	5	5	65	75	86.67%

No.	KF1: User Satisfaction and Overall Usability				KF2: User Experience and Interaction			KF3: Training Effectiveness and Engagement				KF4: System Guidance and Support		KF5: Impact on Confidence and Preparedness		Results		
21	4	3	4	4	4	3	3	3	3	3	3	3	3	2	2	47	75	62.67%
22	5	5	3	5	4	5	5	4	5	4	4	3	4	5	5	66	75	88.00%
23	5	3	3	5	5	2	4	5	4	5	5	4	5	4	3	62	75	82.67%
24	5	5	5	4	5	3	5	5	5	3	4	5	3	3	3	63	75	84.00%
25	4	4	4	5	3	5	4	5	4	5	5	4	2	4	2	60	75	80.00%
26	4	4	4	3	5	4	2	4	4	4	4	4	3	3	1	53	75	70.67%
27	4	5	4	3	4	4	4	5	5	5	5	5	4	5	5	67	75	89.33%
28	5	4	3	4	5	3	3	4	2	5	4	4	4	4	4	58	75	77.33%
29	3	5	5	5	5	3	5	5	3	5	5	5	3	3	5	65	75	86.67%
30	4	3	4	4	4	4	4	4	5	4	4	4	5	5	5	63	75	84.00%
31	5	4	5	5	2	5	5	3	4	5	3	3	5	4	4	62	75	82.67%
	126	126	124	132	132	124	124	122	128	129	123	124	121	116	120			
	155	155	155	155	155	155	155	155	155	155	155	155	155	155	155			
Mean	4.1	4.0	4.0	4.3	4.3	4.0	4.0	4.1	4.1	4.2	4.0	4.0	3.7	3.7	3.9			
% age	81%	81%	80%	85%	85%	80%	80%	79%	83%	83%	79%	80%	78%	75%	77%			
	620				465			620				310		310				
	0.819354839				0.817204301			0.809677419				0.790322581		0.761290323				

Table A-3. Results of novice participants correspond to each key factor.

Welding	Scaffolding	Novice	KF1: User Satisfaction and Overall Usability				KF2: User Experience and Interaction			KF3: Training Effectiveness and Engagement				KF4: System Guidance and Support		KF5: Impact on Confidence and Preparedness				
		o	4	3	4	5	3	5	4	3	5	3	4	3	4	4	5	59	75	78.67%
		o	5	4	4	5	4	5	3	4	5	4	5	5	3	5	5	66	75	88.00%
		o	5	5	3	5	5	5	5	4	5	5	5	5	5	3	5	70	75	93.33%
	o	o	4	5	5	4	5	4	5	5	4	5	4	5	4	5	4	68	75	90.67%
		o	5	5	4	5	5	4	5	3	5	5	3	5	3	5	3	65	75	86.67%
		o	5	5	4	4	5	4	4	4	5	5	5	5	5	5	5	70	75	93.33%
		o	4	4	3	5	4	5	4	5	5	4	5	3	5	4	5	65	75	86.67%
	o	o	4	5	5	4	5	4	4	4	4	5	5	5	5	3	4	66	75	88.00%
		o	4	3	5	4	4	4	5	4	5	5	5	4	5	5	3	65	75	86.67%
	o	o	3	4	4	5	5	5	5	3	4	3	5	4	5	5	5	65	75	86.67%
	o	o	5	5	3	5	4	5	5	4	5	4	4	3	4	5	5	66	75	88.00%
o		o	4	5	4	3	4	4	4	5	5	5	5	5	4	5	5	67	75	89.33%
		o	3	5	5	5	5	3	5	5	3	5	5	5	3	3	5	65	75	86.67%
	o	o	4	3	4	4	4	4	4	4	5	4	4	4	5	5	5	63	75	84.00%
o		o	5	4	5	5	2	5	5	3	4	5	3	3	5	4	4	62	75	82.67%
			64	65	62	68	64	66	67	60	69	67	67	64	65	66	68			
			300				225			300				150		150				
			0.863333333				0.875555556			0.876666667				0.86		0.893333333				
			86.3%				87.6%			87.7%				86.0%		89.3%				

Table A-4. Results of beginner participants correspond to each key factor.

Electrical	Welding	Scaffolding	Beginner	KF1: User Satisfaction and Overall Usability				KF2: User Experience and Interaction			KF3: Training Effectiveness and Engagement				KF4: System Guidance and Support		KF5: Impact on Confidence and Preparedness				
	o		o	3	3	5	5	3	4	3	4	5	3	3	3	5	4	4	57	75	76.00%
	o		o	4	5	5	5	5	4	2	5	4	5	2	5	3	3	5	62	75	82.67%
	o		o	4	4	4	5	4	5	3	5	5	4	3	4	3	3	3	59	75	78.67%
		o	o	5	5	3	4	5	3	3	3	3	3	3	5	4	4	4	57	75	76.00%
o			o	4	4	2	5	4	3	5	5	5	4	4	4	4	2	3	58	75	77.33%
o			o	3	4	4	3	5	4	3	4	3	5	4	3	4	3	4	56	75	74.67%
	o		o	3	4	5	3	5	4	4	2	5	3	3	3	3	3	4	54	75	72.00%
	o		o	5	3	3	5	5	2	4	5	4	5	5	4	5	4	3	62	75	82.67%
o			o	5	5	5	4	5	3	5	5	5	3	4	5	3	3	3	63	75	84.00%
		o	o	4	4	4	5	3	5	4	5	4	5	5	4	2	4	2	60	75	80.00%
o			o	5	4	3	4	5	3	3	4	2	5	4	4	4	4	4	58	75	77.33%
				45	45	43	48	49	40	39	47	45	45	40	44	40	37	39			
				220				165			220				110		110				
				0.822727273				0.775757576			0.804545455				0.763636364		0.690909091				
				82%				78%			80%				76%		69%				

Table A-5. Results of expert participants correspond to each key factor.

Electrical	Welding	Scaffolding	Expert	KF1: User Satisfaction and Overall Usability				KF2: User Experience and Interaction			KF3: Training Effectiveness and Engagement				KF4: System Guidance and Support		KF5: Impact on Confidence and Preparedness				
		o	o	4	4	4	2	4	3	4	2	2	4	3	3	4	3	3	49	75	65.33%
	o		o	2	2	3	4	2	4	5	3	3	2	4	2	4	3	3	46	75	61.33%
	o		o	3	3	4	3	4	4	4	3	2	4	2	4	2	2	4	48	75	64.00%
o			o	4	3	4	4	4	3	3	3	3	3	3	3	3	2	2	47	75	62.67%
o			o	4	4	4	3	5	4	2	4	4	4	4	4	3	3	1	53	75	70.67%
				17	16	19	16	19	18	18	15	14	17	16	16	16	13	13			
				100				75			100				50		50				
				0.68				0.733333333			0.62				0.64		0.52				
				68%				73%			62%				64%		52%				

국문 초록

대화형 AI 기반 가상현실을 활용한 맞춤형 건설 안전 교육

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건설 안전 교육은 현장에서 발생할 수 있는 잠재적 위험을 완화하기 위해 근로자들에게 필수적인 지식과 기술을 제공하는 데 매우 중요하다. 전통적인 일률적 교육 방식은 현장에서 직면하는 다양한 도전에 효과적으로 대비함에 있어 종종 불충분하다는 것이 입증되어왔다. 본 연구는 건설 산업을 위한 개인 맞춤형 안전 교육을 향상시키기 위해 가상 현실(VR) 기술과 인공지능(AI), 특히 대형 언어 모델(LLM)의 통합을 탐구한다. 제안된 시스템은 몰입형 VR 환경과 LLM 기반 지능형 안전 교육 도구(iSafeTrainer)를 결합하여 전기, 비계, 용접 등의 관련 직무에 대해 초보자, 중급자, 전문가 수준의 지식의 지식

수준을 갖춘 교육생들이 대화형 상호작용에 참여할 수 있도록 한다. 시스템은 이러한 개인 맞춤형 접근 방식을 통해 건설 근로자들의 안전 지식을 향상시키는 것을 목표로 한다.

iSafeTrainer 기반 VR 시스템의 효과를 평가하기 위해 설문조사에 기반한 실험 연구가 수행되었다. 참가자들은 VR 시스템과 상호작용하고 LLM 기반 iSafeTrainer 대화하며 구조적 및 비구조적 질문에 대한 피드백을 제공받았다. 제안된 시스템의 평가는 1) 사용자 만족도 및 전반적인 사용성, 2) 사용자 경험 및 상호작용, 3) 교육 효과 및 참여도, 4) 시스템 안내 및 지원, 5) 자신감 및 준비도에 미치는 영향의 5 가지 주요 요소를 사용하여 이루어졌다. 평가 결과 본 시스템을 통한 안전교육의 효과적인 결과를 나타냈으며, 참가자들은 특히 맞춤형 콘텐츠 제공, 몰입형 가상 환경, AI 기반 iSafeTrainer의 실시간 안내와 피드백 측면에서 전통적인 방법 대비 iSafeTrainer를 높이 평가하였다.

본 연구는 VR과 AI 기술 간의 통합이 개인 맞춤형, 상호작용형, 몰입형 학습 경험에 있어 가지는 잠재력을 시연함으로써 건설 안전 교육의 발전에 기여한다. iSafeTrainer 시스템은 개인의 필요에 맞춰 교육 콘텐츠를 동적으로 조정하는 것이 가능함을 보여줌에 따라, 기존 안전 교육 접근 방식의 주요한 격차를 해소할 수 있다.

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